Fact ledger — expert review sheet

681 facts · single source of truth for notes, MCQs and video scripts. Tick the last column (✓ / ✗ + note). ⚠ = volatile fact (drug, guideline, number) — check against current guidance.

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F001	Innate immunity gives continuous barrier defense and reacts to microbes in the first hours or days of infection, before adaptive responses develop.	Abbas 11e p.3	*“provides continuous barrier defenses to prevent infection and also reacts to invading microbes in the first few hours or days after infection”*		
M1-F002	The innate immune system is fully functional before an infection occurs.	Abbas 11e p.3	*“The innate immune system is fully functional in individuals even before an infection occurs”*		
M1-F003	Adaptive immunity is stimulated by exposure to infectious agents and increases in magnitude and defensive capability with repeated exposure.	Abbas 11e p.3	*“increase in magnitude and defensive capabilities with repeated exposures to a particular microbe”*		
M1-F004	The innate response gives early danger signals that stimulate adaptive responses, and adaptive responses often enhance innate protective mechanisms.	Abbas 11e p.3	*“The innate immune response to microbes provides early danger signals that stimulate adaptive immune responses.”*		
M1-F005	The kinetics of innate and adaptive responses are approximations and vary between infections.	Abbas 11e p.3	*“The kinetics of the innate and adaptive immune responses are approximations and may vary in different infections.”*		
M1-F006	Innate immunity is specific for molecules shared by groups of related microbes and for molecules from damaged host cells; adaptive immunity is specific for many microbial and nonmicrobial antigens.	Abbas 11e p.4	*“For molecules shared by groups of related microbes and molecules produced by damaged host cells”*		
M1-F007	Innate diversity is low because germline genes encode the recognition molecules; adaptive diversity is very high because of somatic recombination of gene segments.	Abbas 11e p.4	*“Low; recognition molecules encoded by inherited (germline) genes”*		
M1-F008	Innate immunity has limited memory; adaptive immunity has memory.	Abbas 11e p.4	*“Memory Limited Yes”*		
M1-F009	Both innate and adaptive immunity show nonreactivity to self.	Abbas 11e p.4	*“Nonreactivity to self Yes Yes”*		
M1-F010	Innate components are epithelia, antimicrobial molecules, complement, lectins, phagocytes, dendritic cells, NK cells, mast cells, eosinophils and ILCs; adaptive components are B and T lymphocytes and antibodies.	Abbas 11e p.4	*“Phagocytes (macrophages, neutrophils), dendritic cells, natural killer cells, mast cells, eosinophils, innate lymphoid cells”*		
M1-F011	Innate responses to repeated exposure to the same microbe are usually the same in kinetics, magnitude and quality as the first response.	Abbas 11e p.4	*“the kinetics, magnitude, and qualities of innate responses to repeated exposures of the same microbe are usually the*		

ID	Fact	Source	Supporting quote	Vol.	Verified
			same as after the first exposure”		
M1-F012	Dendritic cells, some macrophages and mast cells are resident in most tissues and act as sentinels.	Abbas 11e p.4	*“Innate immune cells, including DCs, some macrophages, and mast cells, are resident in most tissues, and they function as sentinels”*		
M1-F013	Innate immunity combats microbes by inflammation (recruitment of leukocytes) and by blocking viral replication or killing virus-infected cells.	Abbas 11e p.4	*“by recruiting phagocytes and other leukocytes that destroy the microbes, in the process called inflammation”*		
M1-F014	Lymphocyte activation and differentiation take several days, so the adaptive response is slow and innate immunity must protect first.	Abbas 11e p.10	*“take several days, which explains why the adaptive response is slow to develop and innate immunity has to provide protection initially”*		
M1-F015	Secondary adaptive responses are usually more rapid, greater in magnitude and often qualitatively different from the primary response.	Abbas 11e p.6	*“more rapid, greater in magnitude, and often qualitatively different from the first, or primary, immune response to that antigen”*		
M1-F016	Innate immunity recognizes about 1000 products of microbes and damaged cells; adaptive immunity can recognize millions of antigens.	Abbas 11e p.63	*“the innate immune system recognizes only about 1000 products of microbes and damaged cells”*		
M1-F017	Innate immunity uses germline-encoded invariant receptors; adaptive immunity uses diverse receptors from somatically rearranged genes.	Abbas 11e p.63	*“The innate immune system uses germline-encoded invariant receptors to recognize microbial and other products.”*		
M1-F018	Microbial substances that stimulate innate immunity are often shared by classes of microbes and are called PAMPs.	Abbas 11e p.63	*“are often shared by classes of microbes and are called pathogen-associated molecular patterns (PAMPs)”*		
M1-F019	DAMPs are endogenous molecules released from damaged or dying cells, including cells injured by sterile insults such as burns or trauma.	Abbas 11e p.63	*“These substances are called damage-associated molecular patterns (DAMPs)”*		
M1-F020	Cells that die by apoptosis generally do not release DAMPs.	Abbas 11e p.63	*“DAMPs are generally not released from cells dying by apoptosis.”*		
M1-F021	Innate immunity recognizes microbial products that are often essential for microbial survival, so microbes usually cannot evade it by losing them.	Abbas 11e p.63	*“microbes usually cannot evade innate immunity by eliminating molecules that may be recognized by the host”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F022	Some cytokines released from damaged cells are DAMPs called alarmins.	Abbas 11e p.63	*“are sometimes called alarmins because their presence outside cells sends an alarm to the immune system”*		
M1-F023	Innate immunity uses fewer than 100 types of invariant receptors; adaptive immunity uses only two receptor types (Ig and TCR) with millions of variants.	Abbas 11e p.64	*“<100 different types of invariant receptors”*		
M1-F024	Innate receptors are nonclonal (identical on all cells of a lineage); adaptive receptors are clonal.	Abbas 11e p.64	*“Nonclonal: identical receptors on all cells of the same lineage”*		
M1-F025	Adaptive self-nonreactivity depends on elimination or inactivation of self-reactive lymphocytes and may be imperfect, giving rise to autoimmunity.	Abbas 11e p.64	*“based on elimination or inactivation of self-reactive lymphocytes; may be imperfect (giving rise to autoimmunity)”*		
M1-F026	Innate immunity ignores healthy cells because normal cells lack most ligands, receptors sit in separate compartments, and regulatory proteins block activation.	Abbas 11e p.65	*“normal cells do not produce ligands for most innate immune receptors”*		
M1-F027	Defects in innate components are very rare but increase susceptibility to infection even when adaptive immunity is intact.	Janeway 8e p.37	*“defects in its components, which are very rare, can lead to increased susceptibility to infection, even in the presence of an intact adaptive immune system”*		
M1-F028	The main interfaces with the environment are the skin and the mucosal surfaces of the gastrointestinal, respiratory and genitourinary tracts.	Abbas 11e p.79	*“the skin and the mucosal surfaces of the gastrointestinal, respiratory, and genitourinary tracts”*		
M1-F029	Loss of epithelial integrity predisposes to infection, and infection of barrier epithelial cells is a frequent first event in barrier breakdown.	Abbas 11e p.79-80	*“Loss of the integrity of these epithelial layers by trauma or other reasons predisposes an individual to infections”*		
M1-F030	Epithelial cells form tight junctions that block the passage of microbes between cells.	Abbas 11e p.80	*“The epithelial cells form tight junctions with one another, blocking the passage of microbes between the cells.”*		
M1-F031	The outer keratin layer of skin blocks microbial penetration, and skin infections usually occur after a barrier breach.	Abbas 11e p.80	*“cutaneous infections generally occur when there is a barrier breach”*		
M1-F032	Respiratory, gastrointestinal and genitourinary epithelia produce mucus containing mucins, which physically impairs microbial invasion.	Abbas 11e p.80	*“Mucus, a viscous secretion containing glycoproteins called mucins, is produced by respiratory, gastrointestinal, and urogenital epithelial cells”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F033	Ciliary action in the bronchial tree and peristalsis in the gut help to eliminate microbes.	Abbas 11e p.80	*“ciliary action in the bronchial tree and by peristalsis in the gut, which facilitate elimination of microbes”*		
M1-F034	In cystic fibrosis, thick mucus or impaired ciliary movement leads to lung infections by bacteria that colonize the epithelial surface but do not cross it.	Janeway 8e p.44	*“Such individuals frequently develop lung infections caused by bacteria that colonize the epithelial surface but do not cross it.”*		
M1-F035	Failure of peristalsis is typically accompanied by overgrowth of pathogenic bacteria in the gut lumen.	Janeway 8e p.44	*“Failure of peristalsis is typically accompanied by the overgrowth of pathogenic bacteria within the lumen of the gut.”*		
M1-F036	Surface epithelia provide mechanical, chemical and microbiological barriers to infection.	Janeway 8e p.43	*“Surface epithelia provide mechanical, chemical, and microbiological barriers to infection.”*		
M1-F037	Phagocytes lie beneath all epithelial surfaces and engulf invading microorganisms.	Janeway 8e p.38	*“the phagocytes that lie beneath all epithelial surfaces and that engulf and digest invading microorganisms”*		
M1-F038	Stomach acid, digestive enzymes, bile salts, fatty acids and lysolipids form a chemical barrier in the upper gastrointestinal tract.	Janeway 8e p.44	*“the acid pH of the stomach and the digestive enzymes, bile salts, fatty acids, and lysolipids present in the upper gastrointestinal tract”*		
M1-F039	Lysozyme (a glycosidase) and secretory phospholipase A2 are secreted in tears and saliva and by phagocytes; Paneth cells also make both.	Janeway 8e p.45	*“secreted in tears and saliva and by phagocytes. Lysozyme is a glycosidase ... Lysozyme is also produced by Paneth cells”*		
M1-F040	Lysozyme cleaves peptidoglycan and is more effective against Gram-positive than Gram-negative bacteria, whose LPS outer layer covers the peptidoglycan.	Janeway 8e p.45	*“is more effective against Gram-positive bacteria, in which the peptidoglycan cell wall is exposed”*		
M1-F041	Three important classes of antimicrobial peptides in mammals are defensins, cathelicidins and histatins.	Janeway 8e p.45	*“Three important classes of antimicrobial peptides in mammals are defensins, cathelicidins, and histatins.”*		
M1-F042	Defensins are cationic and hydrophobic peptides of 29 to 34 amino acids with three disulfide bonds; humans have alpha and beta families.	Abbas 11e p.80	*“Defensins are peptides, 29 to 34 amino acids long, that contain both cationic and hydrophobic regions and three intrachain disulfide bonds”*		
M1-F043	Defensins act within minutes to disrupt membranes of bacteria, fungi and some enveloped viruses.	Janeway 8e p.45	*“Defensins act within minutes to disrupt the cell membranes of bacteria, fungi, and the membrane envelopes of some*		

ID	Fact	Source	Supporting quote	Vol.	Verified
			viruses."		
M1-F044	Mucosal epithelial cells and granule-containing leukocytes (neutrophils, NK cells, CTLs) produce defensins.	Abbas 11e p.80	*"produced by epithelial cells of mucosal surfaces and by granule-containing leukocytes, including neutrophils, NK cells, and cytotoxic T lymphocytes"*		
M1-F045	Paneth cells in the crypts of the small bowel are a major source of alpha-defensins.	Abbas 11e p.80	*"Paneth cells within the crypts of the small bowel are a major producer of α-defensins."*		
M1-F046	Beta-defensins are generally made in response to microbial products, mainly by epithelia of the respiratory and urogenital tracts, skin and tongue.	Janeway 8e p.46	*"are generally produced specifically in response to the presence of microbial products"*		
M1-F047	The single human theta-defensin gene is inactivated by a mutation.	Janeway 8e p.46	*"the single human 0-defensin gene has been inactivated by a mutation"*		
M1-F048	Neutrophils and barrier epithelia of skin, gut and airway make cathelicidin as an 18-kD precursor that is cleaved into two active peptides.	Abbas 11e p.80	*"is synthesized as an 18-kD precursor protein that is secreted and then proteolytically cleaved into two peptides"*		
M1-F049	The C-terminal cathelicidin fragment LL-37 binds and neutralizes LPS.	Abbas 11e p.80	*"The C-terminal fragment, called LL-37, can bind and neutralize LPS"*		
M1-F050	DNA-bound LL-37 activates plasmacytoid dendritic cells to secrete type I IFNs, and LL-37 may be an autoantigen in psoriasis.	Abbas 11e p.80	*"LL-37 may also be an autoantigen in human autoimmune diseases such as psoriasis."*		
M1-F051	Neutrophils store cathelicidin propeptides in secondary granules, and elastase from primary granules activates them in the phagosome.	Janeway 8e p.46	*"the cathelicidin propeptides are stored in the secondary granules"*		
M1-F052	Parotid, sublingual and submandibular glands make histatins, which act against Cryptococcus neoformans and Candida albicans.	Janeway 8e p.46	*"are active against pathogenic fungi such as Cryptococcus neoformans and Candida albicans"*		
M1-F053	Antimicrobial peptides are usually made as inactive proproteins that need proteolytic cleavage for activity.	Janeway 8e p.47	*"Antimicrobial peptides are usually made as an inactive proprotein that requires a proteolytic step to complete its activation"*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F054	Barrier epithelia contain intraepithelial T lymphocytes (alpha-beta or gamma-delta TCR) with limited receptor diversity.	Abbas 11e p.80	*“A characteristic of intraepithelial T cells is the limited diversity of their antigen receptors”*		
M1-F055	Commensal microbiota compete with pathogens for nutrients and for attachment sites on epithelial cells.	Janeway 8e p.44	*“compete with pathogenic microorganisms for nutrients and for attachment sites on epithelial cells”*		
M1-F056	Microbiota produce antimicrobial substances, such as lactic acid from vaginal lactobacilli and bacteriocins.	Janeway 8e p.44	*“such as the lactic acid produced by vaginal lactobacilli, some strains of which also produce antimicrobial peptides (bacteriocins)”*		
M1-F057	Commensals strengthen the barrier by stimulating epithelial cells to make antimicrobial peptides.	Janeway 8e p.44	*“by stimulating the epithelial cells to produce antimicrobial peptides”*		
M1-F058	When antibiotics kill commensals, pathogens frequently replace them and cause disease.	Janeway 8e p.44	*“When commensal microorganisms are killed by antibiotic treatment, pathogens frequently replace them and cause disease”*		
M1-F059	Commensals can cause disease if their growth is not controlled or if the immune system is compromised.	Janeway 8e p.44	*“commensal microbes themselves can cause disease if their growth is not kept in check or if the immune system is compromised”*		
M1-F060	The response to a new pathogen has three phases: immediate innate (0-4 h), early induced innate (4-96 h) and adaptive (>96 h).	Janeway 8e p.37	*“(immediate: 0-4 hours) ... (early: 4-96 hours) ... (late: >96 hours)”*		
M1-F061	Immediate defenses are preformed soluble molecules such as lysozyme, defensins and complement.	Janeway 8e p.37	*“These first defenses include several classes of preformed soluble molecules present in blood, extracellular fluid, and epithelial secretions”*		
M1-F062	In the second phase, innate cells recognize PAMPs and become activated.	Janeway 8e p.37	*“these innate immune cells sense the presence of a pathogen by recognizing molecules typical of a microbe”*		
M1-F063	Neither soluble nor cellular innate components generate long-term protective memory.	Janeway 8e p.37	*“neither the soluble nor the cellular components of innate immunity generate long-term protective immunological memory”*		
M1-F064	Adaptive immunity is late because rare specific B and T cells must first expand clonally; effector mechanisms are similar in each phase.	Janeway 8e p.37	*“The effector mechanisms that remove the infectious agent are similar or identical in each phase.”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F065	In the induced phase of innate immunity, an inflammatory response often first counters the spread of a pathogen.	Janeway 8e p.42	*“In the second, induced, phase of the innate immune response, the spread of a pathogen is often initially countered by an inflammatory response”*		
M1-F066	Induced innate responses act over several days while dendritic cells carry antigen to local lymphoid tissue to start the adaptive response.	Janeway 8e p.43	*“The induced responses of innate immunity act over several days”*		
M1-F067	Innate defenses act within minutes, and an adaptive response is needed only when innate defenses are bypassed, evaded or overwhelmed.	Janeway 8e p.38	*“Only when the innate host defenses have been bypassed, evaded, or overwhelmed is an adaptive immune response required.”*		
M1-F068	Innate immunity keeps pathogen numbers in check while the adaptive response develops.	Janeway 8e p.38	*“the innate immune system performs a valuable delaying function, keeping pathogen numbers in check”*		
M1-F069	Acute inflammation is the accumulation of leukocytes, plasma proteins and fluid from the blood at an extravascular site of infection or injury.	Abbas 11e p.89	*“accumulation of leukocytes, plasma proteins, and fluid derived from the blood at an extravascular tissue site of infection or injury”*		
M1-F070	Innate immunity protects by inducing inflammation, inducing antiviral defense and stimulating adaptive immunity.	Abbas 11e p.89	*“are by inducing inflammation, inducing antiviral defense, and stimulating adaptive immunity”*		
M1-F071	Tissue sentinel cells (macrophages, DCs, mast cells) and epithelial cells start acute inflammation after recognition of PAMPs or DAMPs.	Abbas 11e p.90	*“mainly macrophages, DCs, and mast cells, as well as by epithelial barrier cells”*		
M1-F072	Inflammation delivers effector molecules and cells, induces local clotting, and promotes tissue repair.	Janeway 8e p.82-83	*“Inflammation has three essential roles in combating infection.”*		
M1-F073	Inflammation is characterized by pain, redness, heat and swelling at the site of infection.	Janeway 8e p.83	*“Inflammatory responses are operationally characterized by pain, redness, heat, and swelling at the site of an infection”*		
M1-F074	Vasodilation increases local blood flow (heat and redness) and slows flow along vessel walls.	Janeway 8e p.83	*“leading to increased local blood flow—hence the heat and redness”*		
M1-F075	Activated endothelium expresses adhesion molecules, and leukocytes attach and migrate into tissue (extravasation).	Janeway 8e p.83	*“allows leukocytes to attach to the endothelium and migrate into the tissues, a process known as extravasation”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F076	Increased vascular permeability causes swelling (edema) and pain and brings plasma proteins such as complement and MBL into tissue.	Janeway 8e p.83	*“accounts for the swelling, or edema, and pain”*		
M1-F077	Clotting in microvessels at the infection site prevents spread of the pathogen through the blood.	Janeway 8e p.83	*“prevents the spread of the pathogen via the blood”*		
M1-F078	Macrophages and neutrophils secrete lipid mediators: prostaglandins, leukotrienes and platelet-activating factor (PAF).	Janeway 8e p.83	*“secrete lipid mediators of inflammation—prostaglandins, leukotrienes, and platelet-activating factor (PAF)”*		
M1-F079	Tissue injury activates the kinin system, which produces bradykinin; bradykinin increases vascular permeability and causes pain.	Janeway 8e p.84	*“Bradykinin causes an increase in vascular permeability that promotes the influx of plasma proteins”*		
M1-F080	C5a increases vascular permeability and activates mast cells to release histamine and TNF.	Janeway 8e p.84	*“C5a also activates local mast cells”*		
M1-F081	Mast cells respond to PAMPs and DAMPs with histamine and prostaglandins, which cause vasodilation and increased capillary permeability.	Abbas 11e p.94	*“Mast cells respond to PAMPs and DAMPs by secreting histamine and prostaglandins that cause vasodilation and increased capillary permeability”*		
M1-F082	TNF and IL-1 induce endothelial E-selectin, ICAM-1 and VCAM-1 and stimulate chemokines such as CXCL8 and CCL2.	Abbas 11e p.94	*“Expression of E-selectin and ligands for integrins, including intercellular adhesion molecule 1 (ICAM-1) and vascular cell adhesion molecule 1 (VCAM-1)”*		
M1-F083	Neutrophils arrive first because they are the most abundant leukocytes and the most rapid responders; monocytes become more numerous over time and may dominate some reactions because they live longer and can proliferate locally.	Abbas 11e p.94	*“may be the dominant population in some reactions because they have longer lifespans than neutrophils and are capable of local proliferation”*		
M1-F084	Acute inflammation develops in minutes to hours and lasts for days.	Abbas 11e p.94	*“Acute inflammation develops in minutes to hours and lasts for days.”*		
M1-F085	Chronic inflammation follows if infection or injury persists; it involves monocytes and lymphocytes, angiogenesis and fibrosis.	Abbas 11e p.95	*“It usually involves recruitment and activation of monocytes and lymphocytes.”*		
M1-F086	TNF, IL-1 and IL-6 are three of the most important proinflammatory cytokines of innate immunity; innate cytokines are made mainly by tissue macrophages and DCs.	Abbas 11e p.90-91	*“Three of the most important proinflammatory cytokines of the innate immune system are TNF, IL-1”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F087	Epithelial IL-25, TSLP and IL-33 stimulate ILC2s, Th2 cells and mast cells, and contribute to allergic disease; IL-33 is an alarmin.	Abbas 11e p.94	*“It is often called an alarmin because it is rapidly released from damaged epithelial cells”*		
M1-F088	Inadequate TNF, for example with TNF-blocking drugs, may cause failure to contain infections.	Abbas 11e p.94	*“a consequence may be failure to contain infections”*		
M1-F089	TNF and IL-1 act on the hypothalamus to cause fever and are called endogenous pyrogens.	Abbas 11e p.96	*“These cytokines are therefore called endogenous pyrogens”*		
M1-F090	Prostaglandin synthesis inhibitors such as aspirin reduce fever by blocking the cytokine effect.	Abbas 11e p.97	*“Prostaglandin synthesis inhibitors, such as aspirin, reduce fever by blocking this action of the cytokines.”*		
M1-F091	Endogenous pyrogens (TNF-alpha, IL-1beta, IL-6) induce cyclooxygenase-2 and prostaglandin E2; PGE2 acts on the hypothalamus to increase heat production and vasoconstriction.	Janeway 8e p.109	*“Endogenous pyrogens cause fever by inducing the synthesis of prostaglandin E2 by the enzyme cyclooxygenase-2”*		
M1-F092	LPS is an exogenous pyrogen that also induces cyclooxygenase-2 directly through TLR4 signaling.	Janeway 8e p.109	*“directly inducing cyclooxygenase-2 as a consequence of signaling through TLR-4”*		
M1-F093	The role of fever in host defense is not well understood.	Abbas 11e p.97	*“The role of fever in host defense is not well understood”*		
M1-F094	Fever is generally beneficial because most pathogens grow better at lower temperatures and adaptive responses are more intense at higher temperatures.	Janeway 8e p.109	*“Fever is generally beneficial to host defense; most pathogens grow better at lower temperatures”*		
M1-F095	TNF, IL-1 and IL-6 act on the bone marrow, with colony-stimulating factors, to cause leukocytosis.	Abbas 11e p.97	*“leading to elevated numbers of these white cells in the blood (leukocytosis)”*		
M1-F096	TNF, IL-1 and IL-6 induce hepatocytes to make acute-phase proteins such as CRP, SAP and fibrinogen; raised CRP is a clinical sign of inflammation.	Abbas 11e p.97	*“Elevated plasma CRP is used clinically as a sign of an inflammatory process.”*		
M1-F097	Plasma CRP can increase up to 1000-fold in infection because IL-6, IL-1 and TNF increase its synthesis by liver cells.	Abbas 11e p.88	*“can increase up to 1000-fold during infections and in response to other inflammatory stimuli”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F098	CRP, SAP and PTX3 activate the classical complement pathway by binding C1q.	Abbas 11e p.88	*“CRP, SAP, and PTX3 all activate the classical pathway of complement by binding C1q.”*		
M1-F099	PTX3 is made by DCs, macrophages and endothelial cells and is also stored in neutrophil granules.	Abbas 11e p.88	*“PTX3 is also stored in neutrophil granules and released as neutrophils die”*		
M1-F100	In the acute-phase response, blood levels of some proteins go down while others increase markedly.	Janeway 8e p.109	*“blood levels of some proteins go down, whereas levels of others increase markedly”*		
M1-F101	CRP is a pentraxin that binds phosphocholine on bacterial and fungal surfaces but not in host membranes; it opsonizes and binds C1q.	Janeway 8e p.110	*“it binds to the phosphocholine portion of certain bacterial and fungal cell-wall lipopolysaccharides”*		
M1-F102	MBL is low in healthy blood and increases during the acute-phase response.	Janeway 8e p.110	*“MBL is present in low levels in the blood of healthy individuals”*		
M1-F103	Surfactant proteins SP-A and SP-D increase in the acute-phase response and promote phagocytosis of pathogens such as Pneumocystis jirovecii.	Janeway 8e p.110	*“important in promoting the phagocytosis of opportunistic respiratory pathogens such as Pneumocystis jirovecii”*		
M1-F104	Within a day or two, acute-phase proteins give antibody-like functions, but they have no structural diversity and are induced nonspecifically.	Janeway 8e p.110	*“acute-phase proteins have no structural diversity”*		
M1-F105	Janeway lists serum amyloid P as an acute-phase protein in mice but not in humans.	Janeway 8e p.111	*“serum amyloid protein (SAP) (in mice but not humans)”*		
M1-F106	Sepsis shows fever, fast heart and respiratory rates, metabolic abnormalities and mental disturbance, often started by LPS or lipoteichoic acid.	Abbas 11e p.97	*“clinically characterized by fever, fast heart and respiratory rates, metabolic abnormalities, and mental disturbances”*		
M1-F107	In septic shock, high TNF causes vascular collapse and disseminated intravascular coagulation.	Abbas 11e p.97	*“there is vascular collapse and disseminated intravascular coagulation, caused by the effects of high doses of TNF”*		
M1-F108	Systemic TNF reduces myocardial contractility and vascular tone, causes intravascular thrombosis and, if prolonged, cachexia.	Abbas 11e p.97	*“TNF inhibits myocardial contractility and vascular smooth muscle tone”*		
M1-F109	Clinical trials of anti-TNF antibodies or soluble TNF receptors have not shown benefit in sepsis.	Abbas 11e p.97	*“have not shown a benefit in patients with sepsis”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F110	A septic-shock-like syndrome after burns, trauma or pancreatitis is called the systemic inflammatory response syndrome (SIRS).	Abbas 11e p.97	*“This has been called the systemic inflammatory response syndrome (SIRS).”*		
M1-F111	TNF blockade can reactivate tuberculosis in apparently well patients with previous infection, which shows the role of TNF in keeping infection local.	Janeway 8e p.109	*“these treatments have been found to reactivate tuberculosis in some apparently well patients with evidence of previous infection”*		
M1-F112	Lymphocyte activation needs two signals: antigen (signal 1) and molecules made during innate responses (signal 2).	Abbas 11e p.100	*“The activation of lymphocytes requires two distinct signals, the first being antigen”*		
M1-F113	Signal 1 makes the response specific; signal 2 makes sure adaptive responses start for dangerous infections and not for harmless or self antigens.	Abbas 11e p.100	*“ensures that adaptive immune responses are induced when there is a dangerous infection and not when lymphocytes recognize harmless antigens”*		
M1-F114	Second signals are costimulators (T cells), cytokines (T and B cells) and complement breakdown products (B cells).	Abbas 11e p.100	*“costimulators (for T cells), cytokines (for both T and B cells), and complement breakdown products (for B cells)”*		
M1-F115	Second signals increase the magnitude of the adaptive response and also influence its type.	Abbas 11e p.100	*“not only enhance the magnitude of the subsequent adaptive immune response but also influence the nature of the adaptive response”*		
M1-F116	IL-12 drives Th1; IL-1, IL-6 and IL-23 drive Th17; IL-25, IL-33 and TSLP drive Th2; IL-15 supports memory CD8+ T cells; IL-6 supports plasma cells.	Abbas 11e p.101	*“IL-25, IL-33, and TSLP stimulate the differentiation of naive CD4+ T cells to the Th2 subset of effector cells”*		
M1-F117	Blood-borne microbes activate the alternative complement pathway, and complement products enhance antibody production by B cells.	Abbas 11e p.101	*“some of the proteolytic products of complement activation enhance the production of antibodies by B lymphocytes”*		
M1-F118	Adjuvants work mostly by stimulating innate responses; aluminium salts are the most common human vaccine adjuvant and may act by inflammasome activation.	Abbas 11e p.101	*“Aluminum, in the form of various salts, is the most commonly used adjuvant in human vaccines and may work by causing inflammasome activation”*		
M1-F119	CpG, the TLR9 ligand, is used as an adjuvant in some hepatitis B virus vaccines.	Abbas 11e p.101	*“CpG, the ligand for TLR9, is used in some hepatitis B virus vaccines”*	⚠	
M1-F120	Adjuvants make DCs express more MHC molecules (signal 1), costimulators (signal 2) and cytokines, and migrate to lymph nodes.	Abbas 11e p.101	*“increase the expression of costimulators (signal 2) and cytokines needed for T-cell activation”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M1-F121	PAMP-activated DCs express costimulators and direct naive T cells to effector types such as Th1 or Th17; DCs bridge innate and adaptive immunity.	Abbas 11e p.81	*“DCs activated by microbial PAMPs also express membrane molecules called costimulators that promote T-cell responses.”*		
M1-F122	TNF-alpha stimulates DC migration from tissues to lymph nodes and maturation into highly costimulatory antigen-presenting cells.	Janeway 8e p.111	*“TNF-a also has a role in stimulating the migration of dendritic cells from their sites in peripheral tissues to the lymph node”*		
M1-F123	IgG-mediated phagocytosis shows that an adaptive product (antibody) engages innate effector cells.	Abbas 11e p.95	*“Antibody-dependent phagocytosis illustrates a link between innate and adaptive immunity”*		
M1-F124	IL-10 is made by and inhibits macrophages and DCs (negative feedback); loss-of-function IL-10 receptor mutations cause severe infantile colitis.	Abbas 11e p.101	*“Loss-of-function mutations in the IL-10 receptor result in severe colitis that develops in infancy.”*		
M1-F125	Antagonists of TNF, IL-1, IL-6 and IL-12 or their receptors are used to reduce inflammation in rheumatoid arthritis, inflammatory bowel disease and psoriasis.	Abbas 11e p.97	*“Antagonists against all of these cytokines or their receptors are used clinically to reduce inflammation”*		
M1-F126	In the airways, goblet cells secrete mucus, and beating cilia move the mucus outward to trap and eject pathogens.	Janeway 8e p.48	*“Beating of the cilia moves a continuous stream of mucus secreted by goblet cells (green) outward, trapping and ejecting potential pathogens”*		
M1-F127	The pentraxins include the short pentraxins CRP and SAP and the long pentraxin PTX3.	Abbas 11e p.87	*“include the short pentraxins, C-reactive protein (CRP) and serum amyloid P (SAP), and the long pentraxin PTX3”*		
M2-F001	All blood cells, including granulocytes, monocytes, dendritic cells (DCs), mast cells, lymphocytes and innate lymphoid cells (ILCs), come from one haematopoietic stem cell (HSC) in the bone marrow.	Abbas 11e p.30	*“granulocytes, monocytes, DCs, mast cells, platelets, B and T lymphocytes, and ILCs all originate from a common HSC in the bone marrow”*		
M2-F002	HSCs express CD34 and c-KIT and do not express lineage-specific markers.	Abbas 11e p.30	*“HSCs can be identified by the presence of surface markers, including the proteins CD34 and c-KIT, and the absence of lineage-specific markers”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F003	Common myeloid-megakaryocyte-erythroid progenitors give rise to the granulocytic lineage (neutrophils, eosinophils, basophils) and to monocytes.	Abbas 11e p.32	*“give rise to committed precursors of the erythroid, megakaryocytic, granulocytic, and monocytic lineages”*		
M2-F004	Most DCs arise from a precursor that also gives rise to monocytes.	Abbas 11e p.32	*“most DCs arise from a precursor that also gives rise to monocytes”*		
M2-F005	Mast cell progenitors arise from a common granulocyte/monocyte precursor, leave the bone marrow and mature in peripheral tissues.	Abbas 11e p.32	*“Immature mast cell progenitors arise from a common granulocyte/monocyte precursor, leave the bone marrow, and mature into mast cells in peripheral tissues”*		
M2-F006	The common lymphoid progenitor gives rise to T cells, B cells and a shared precursor of natural killer (NK) cells and ILCs.	Abbas 11e p.30	*“The common lymphoid progenitor in the bone marrow that gives rise to T and B lymphocytes also gives rise to a precursor of both NK”*		
M2-F007	G-CSF and GM-CSF stimulate neutrophil production.	Abbas 11e p.14	*“Production of neutrophils is stimulated by granulocyte colony-stimulating factor (G-CSF) and granulocyte-macrophage colony-stimulating factor (GM-CSF)”*		
M2-F008	M-CSF drives the development of the monocyte-macrophage lineage from bone marrow precursors.	Abbas 11e p.15	*“driven by a cytokine called a monocyte (or macrophage) colony-stimulating factor (M-CSF)”*		
M2-F009	All DC subsets depend on FLT3 ligand (FLT3L) for development and express CD11c.	Abbas 11e p.20	*“dependence on the cytokine FLT3L for their development, expression of the CD11c protein”*		
M2-F010	Basophils mature in the bone marrow from progenitors that are different from mast cell progenitors.	Abbas 11e p.20	*“mature in the bone marrow (from progenitors distinct from those of mast cells), and circulate in the blood”*		
M2-F011	Janeway (8th ed.) states that DCs arise from both myeloid and lymphoid progenitors.	Janeway 8e p.77	*“Dendritic cells arise from both myeloid and lymphoid progenitors within the bone marrow”*		
M2-F012	Normal blood differential: neutrophils 40-60%, eosinophils 1-4%, basophils <1%, monocytes 2-8% of leukocytes.	Abbas 11e p.14	*“Neutrophils 4400 40%–60% Eosinophils 200 1%–4% Basophils 40 <1%”*		
M2-F013	Neutrophils are the most abundant circulating white blood cells and the principal cell of acute inflammation.	Abbas 11e p.14	*“Neutrophils are the most abundant population of circulating white blood cells and the principal cell type in acute inflammatory reactions”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F014	The neutrophil nucleus has three to five lobes, so neutrophils are also called polymorphonuclear leukocytes (PMNs).	Abbas 11e p.14	*“The nucleus is segmented into three to five connected lobules”*		
M2-F015	Most neutrophil granules are specific granules, which contain lysozyme, collagenase and elastase.	Abbas 11e p.14	*“called specific granules, are filled with enzymes, such as lysozyme, collagenase, and elastase”*		
M2-F016	Neutrophil azurophilic granules contain myeloperoxidase, defensins and cathelicidins.	Abbas 11e p.14	*“azurophilic granules ... contain enzymes (e.g., myeloperoxidase) and microbicidal substances, including defensins and cathelicidins”*		
M2-F017	Human neutrophil surface markers include CD11b, CD18, CD15, CD66b and CD10.	Abbas 11e p.15	*“Surface markers (human) CD11b, CD18, CD15, CD66b, CD10”*		
M2-F018	Neutrophils live 1-2 days in tissues, whereas tissue-resident macrophages can live for years.	Abbas 11e p.15	*“Life span in tissues 1–2 days ... Tissue-resident macrophages: years”*		
M2-F019	Neutrophils respond rapidly by activating preformed enzymes; macrophage responses are slower and often need new gene transcription.	Abbas 11e p.15	*“Rapid, short-lived, activation of preformed enzymes ... More prolonged, slower, often dependent on new gene transcription”*		
M2-F020	Neutrophils can extrude nuclear contents as neutrophil extracellular traps (NETs) that immobilize and kill microbes but can also damage healthy tissue.	Abbas 11e p.15	*“extrude their nuclear contents, forming neutrophil extracellular traps (NETs), which serve to immobilize and kill extracellular microbes but also may damage healthy tissues”*		
M2-F021	Classical (inflammatory) monocytes form 90-95% of human blood monocytes; they are rapidly recruited to sites of infection or injury, where they differentiate into macrophages.	Abbas 11e p.15-16	*“Blood monocytes are efficiently recruited into tissue sites of infection or injury, where they differentiate (mature) into macrophages”*		
M2-F022	Most tissue-resident macrophages are long-lived, self-renewing cells that derive from yolk sac or fetal liver precursors, not from bone marrow.	Abbas 11e p.15	*“Most tissue-resident macrophages are long-lived and are derived not from the bone marrow but from yolk sac or fetal liver precursors”*		
M2-F023	Examples of tissue-resident macrophages are Kupffer cells (liver), alveolar macrophages (lung) and microglia (brain).	Abbas 11e p.15	*“Kupffer cells lining the sinusoids in the liver, alveolar macrophages in the lung, and microglial cells in the brain”*		
M2-F024	Janeway (8th ed.) describes tissue macrophages as maturing continuously from blood monocytes.	Janeway 8e p.76	*“Macrophages mature continuously from monocytes that leave the circulation to migrate into tissues throughout the*		

ID	Fact	Source	Supporting quote	Vol.	Verified
			body”		
M2-F025	Human macrophage markers include CD11b, CD14, CD68, CD16, CD64 and CCR5.	Abbas 11e p.15	*“CD11b, CD14, CD68, CD16, CD64, CCR5”*		
M2-F026	Macrophages remove apoptotic cells before they release their contents; this clearance is called efferocytosis.	Abbas 11e p.16	*“This clearance of apoptotic cells by macrophages is called efferocytosis”*		
M2-F027	The inhibitory receptor SIRPalpha on macrophages binds CD47 on healthy cells, a 'don't eat me' signal that blocks phagocytosis.	Abbas 11e p.19	*“an inhibitory receptor on the macrophage called SIRPα, which recognizes CD47, a membrane protein on healthy cells that functions as a “don’t eat me” signal”*		
M2-F028	Classical activation produces microbicidal M1 macrophages; alternative activation produces M2 macrophages that promote tissue remodeling and repair.	Abbas 11e p.19	*“efficient at killing microbes, called classical activation, and these cells are often called M1 macrophages ... alternative activation, and these cells are called M2 macrophages”*		
M2-F029	Macrophages are activated by microbial products such as lipopolysaccharide and by NK cell-derived interferon-gamma (IFN-gamma).	Abbas 11e p.19	*“Macrophages are activated by microbial products such as lipopolysaccharides and by natural killer cell–derived interferon-γ (IFN-γ)”*		
M2-F030	In Abbas Fig. 2.4, cytokines such as IL-4 and IL-13 drive macrophage production of collagen and angiogenic factors, leading to tissue repair and fibrosis.	Abbas 11e p.19	*“Cytokine (e.g., IL-4, IL-13) ... Production of collagen, angiogenic factors ... Tissue repair, fibrosis”*		
M2-F031	Many macrophage subsets do not fall neatly into M1 or M2 categories.	Abbas 11e p.19	*“multiple macrophage subsets exist that do not neatly fall into M1 or M2 categories”*		
M2-F032	Activated macrophages produce cytokines such as TNF, IL-1, IL-6 and IL-12.	Abbas 11e p.19	*“Cytokines (TNF, IL-1, IL-6, IL-12)”*		
M2-F033	DCs are sentinels in epithelia and tissues that capture antigens and carry them to lymph nodes, linking innate and adaptive immunity.	Abbas 11e p.20	*“their ability to capture antigens and take them to lymph nodes where naive T cells circulate ... link innate responses with the development of adaptive”*		
M2-F034	cDC2 cells are the most numerous DCs and present antigens mainly to CD4+ T cells.	Abbas 11e p.21	*“cDC2 cells are the most numerous DCs and are potent at capturing exogenous antigens and inducing CD4+ T-cell responses”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F035	cDC1 cells (IRF8, BATF3; CD141, CLEC9A, XCR1) cross-present antigens to naive CD8+ T cells and produce IL-12 to induce Th1 responses.	Abbas 11e p.21-23	*“The cDC1 subset is specialized to present antigens to naive CD8+ T cells by a process called cross-presentation”*		
M2-F036	Plasmacytoid DCs (pDCs) express BDCA-2 (CD303) and CD123, have high levels of TLR7 and TLR9, use E2-2, and produce type I interferon (IFN).	Abbas 11e p.23	*“BDCA-2 (CD303) BDCA-4 (CD304) CD123 ... High levels of TLR7, TLR9 ... Type I IFN”*		
M2-F037	pDCs are the body's major producers of type I IFNs, which have potent antiviral activity.	Abbas 11e p.22	*“Plasmacytoid DCs are the body’s major producers of type I IFNs, which have potent antiviral activities”*		
M2-F038	pDCs can make up to 1000 times more type I IFN than other cell types.	Janeway 8e p.111	*“make abundant type I interferons (IFN-α and IFN-β)—up to 1000 times more than other cell types”*		
M2-F039	Langerhans cells are epidermal DCs with Birbeck granules and langerin (CD207); they derive from embryonic precursors related to tissue macrophages and present antigens to CD4+ T cells or induce tolerance.	Abbas 11e p.22	*“Langerhans cells may function in the context of skin infections to present antigens to and activate CD4+ T cells, or ... induce tolerance”*		
M2-F040	Follicular dendritic cells are stromal cells that are not bone marrow-derived and do not present protein antigens to T cells.	Abbas 11e p.22	*“follicular dendritic cells (FDCs) have a dendritic morphology but are not derived from bone marrow precursors, do not present protein antigens to T cells”*		
M2-F041	PAMP-activated DCs express costimulators and direct naive T-cell differentiation into effector subsets such as Th1 or Th17 cells.	Abbas 11e p.81	*“DCs activated by microbial PAMPs also express membrane molecules called costimulators ... DCs will direct naive T-cell differentiation into distinct types of effector cells”*		
M2-F042	NK cells form 5-20% of lymphocytes in blood and spleen and are abundant in liver and placenta.	Abbas 11e p.82	*“NK cells constitute 5% to 20% of lymphocytes in the blood and spleen ... they are abundant in the liver and placenta”*		
M2-F043	Blood NK cells are identified as CD56-positive and CD3-negative; most human blood NK cells also express CD16.	Abbas 11e p.82	*“identified in the blood by the expression of CD56 and the absence of the T-cell marker CD3. Most human blood NK cells also express CD16”*		
M2-F044	NK cells kill infected cells without prior activation, using germline-encoded receptors, not clonally distributed antigen receptors.	Abbas 11e p.82	*“NK cells are functionally competent to kill other cells ... without further activation (hence natural) ... use germline DNA–encoded receptors”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F045	NK cells kill infected cells and secrete IFN-gamma, which activates macrophages to destroy phagocytosed microbes.	Abbas 11e p.82	*“The effector functions of NK cells are to kill infected cells and to produce IFN-γ, which activates macrophages to destroy phagocytosed microbes”*		
M2-F046	In NK granules, perforin helps granzymes enter the target cell, and granzymes trigger apoptosis.	Abbas 11e p.82	*“perforin, facilitates the entry of other granule proteins, called granzymes ... cause death of the target cells by apoptosis”*		
M2-F047	NK cells kill virus-infected cells early, before antigen-specific CTLs become fully active after 5-7 days.	Abbas 11e p.82	*“they kill infected cells before antigen-specific CTLs can become fully active, which usually takes 5 to 7 days”*		
M2-F048	IL-12, IL-15, IL-18 and type I IFNs enhance NK cytotoxicity and IFN-gamma secretion; IL-15 is an NK growth factor.	Abbas 11e p.85	*“The major cytokines of the innate immune system that stimulate NK cell function are IL-12, IL-15, IL-18, and type I IFNs”*		
M2-F049	NK cell function depends on the balance between signals from activating and inhibitory receptors.	Abbas 11e p.83	*“NK cell function is regulated by a balance between signals that are generated by activating and inhibitory receptors”*		
M2-F050	NKG2D is an activating receptor that binds stress-induced MIC-A, MIC-B and ULBP proteins on infected and tumour cells and signals through DAP10.	Abbas 11e p.83	*“NKG2D, which binds proteins structurally similar to MHC-I molecules, including MHC-I-related gene A (MIC-A) and MIC-B, and proteins of the ULBP ... DAP10”*		
M2-F051	Natural cytotoxicity receptors NKp30, NKp44 and NKp46 are activating NK receptors.	Janeway 8e p.117	*“Activating receptors for the recognition of infected cells are the natural cytotoxicity receptors (NCRs) NKp30, NKp44, and NKp46”*		
M2-F052	CD16 (FcgammaRIIIA) is a low-affinity IgG receptor that binds IgG1 and IgG3 and mediates antibody-dependent cell-mediated cytotoxicity (ADCC).	Abbas 11e p.84	*“CD16 (FcγRIIIA), which is a low-affinity receptor for IgG antibodies ... binds to the Fc regions of certain types of antibodies called IgG1 and IgG3”*		
M2-F053	NK inhibitory receptors recognise MHC class I (MHC-I); viruses, cell stress and many cancers reduce MHC-I, and NK activation by cells that lack MHC-I is called recognition of missing self.	Abbas 11e p.85	*“many viruses and other causes of cell stress lead to a loss of cell surface expression of MHC-I ... has been called recognition of missing”*		
M2-F054	Inhibitory KIRs bind classical MHC-I molecules; the inhibitory CD94/NKG2A heterodimer binds HLA-E, which displays peptides from other MHC-I molecules.	Abbas 11e p.85	*“CD94/NKG2A heterodimer, which recognizes a non-classical MHC-I molecule called HLA-E. Interestingly, HLA-E displays peptides derived from other MHC-I molecules”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F055	Inhibitory NK receptors carry ITIMs that recruit tyrosine phosphatases; activating receptors signal through ITAMs.	Abbas 11e p.85	*“Inhibitory receptors of NK cells have immunoreceptor tyrosine-based inhibition motifs (ITIMs) ... recruitment and activation of tyrosine phosphatases”*		
M2-F056	Inhibitory KIRs (KIR-2DL, KIR-3DL) have long cytoplasmic tails with ITIMs; activating KIRs (KIR-2DS, KIR-3DS) have short tails and pair with DAP12.	Janeway 8e p.115-116	*“Inhibitory KIRs have long cytoplasmic tails that contain an immunoreceptor tyrosine-based inhibition motif ... Activating KIR receptors have short cytoplasmic tails”*		
M2-F057	NKG2A contains an ITIM and is inhibitory; NKG2C pairs with DAP12 and is activating.	Janeway 8e p.116	*“NKG2A contains an ITIM and is inhibitory, whereas NKG2C has a charged transmembrane residue, associates with DAP12, and is activating”*		
M2-F058	Interferons increase MHC-I on uninfected cells and so protect them from NK cells.	Janeway 8e p.115	*“interferons, which induce the expression of MHC class I molecules, protect uninfected host cells from NK cells”*		
M2-F059	Patients with numerical or functional NK cell deficiency present with severe herpesvirus and papillomavirus infections.	Abbas 11e p.488	*“present with severe infections with viruses, mainly of the herpesvirus and papillomavirus families”*		
M2-F060	Autosomal dominant GATA2 mutations cause loss of NK cells together with reduced monocytes, DCs and B cells.	Abbas 11e p.488	*“autosomal dominant mutations in the gene encoding the GATA2 transcription factor ... loss of NK cells, as well as decreases in monocytes, dendritic cells (DCs)”*		
M2-F061	Autosomal recessive CD16 mutations cause a functional NK cell deficiency.	Abbas 11e p.488	*“One example of functional NK cell deficiency is autosomal recessive mutations affecting CD16 (FcγRIIIA)”*		
M2-F062	Chediak-Higashi syndrome is an autosomal recessive disorder (LYST mutation) in which NK cell function is impaired because of abnormal cytotoxic granules.	Abbas 11e p.488	*“Chédiak-Higashi syndrome (CHS) is a rare autosomal recessive disorder ... NK cell function in these patients is impaired, probably because of an abnormality in the”*		
M2-F063	Mature mast cells are not found in the circulation; they lie near blood vessels and nerves and beneath epithelia.	Abbas 11e p.465	*“Normally, mature mast cells are not found in the circulation ... predominantly near blood vessels (Fig. 20.2A) and nerves and beneath epithelia”*		
M2-F064	Stem cell factor (SCF) binds c-KIT on mast cell precursors and is essential for mast cell development.	Abbas 11e p.465	*“stem cell factor released by tissue cells, which binds to the c-KIT receptor on the mast cell precursors”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F065	Mast cells live weeks to months; basophils live days; eosinophils live days to weeks.	Abbas 11e p.466	*“Life span Weeks to months Days Days to weeks”*		
M2-F066	Major growth factors: mast cells SCF and IL-3; basophils IL-3; eosinophils IL-5.	Abbas 11e p.466	*“Major growth and differentiation factor (cytokines) Stem cell factor, IL-3 IL-3 IL-5”*		
M2-F067	FcepsilonRI expression is high on mast cells and basophils and low on eosinophils.	Abbas 11e p.466	*“Expression of FcεRI High High Low”*		
M2-F068	FcepsilonRI binds IgE with very high affinity (Kd about 10^-10 M), so most mast cells are coated with IgE even in nonatopic people.	Abbas 11e p.466	*“the majority of mast cells are always coated with IgE, even in nonatopic individuals”*		
M2-F069	FcepsilonRI on mast cells and basophils is an alpha-beta-gamma2 tetramer; on eosinophils it is mainly an alpha-gamma2 trimer at low levels.	Abbas 11e p.467	*“FcεRI on mast cells and basophils is expressed as an αβγ2 tetramer, the receptors on eosinophils are mainly αγ2 trimers and are expressed at low”*		
M2-F070	Mast cells are activated when multivalent antigen cross-links IgE bound to FcepsilonRI.	Abbas 11e p.468	*“Mast cells are activated by cross-linking of FcεRI molecules, which occurs by binding of multivalent antigens to the IgE molecules”*		
M2-F071	Mast cell activation causes degranulation, synthesis of lipid mediators and synthesis of cytokines.	Abbas 11e p.468	*“secretion of preformed granule contents by exocytosis (degranulation), synthesis and secretion of lipid mediators, and synthesis and secretion of cytokines”*		
M2-F072	Mast cell and basophil granules store histamine, proteoglycans (heparin and/or chondroitin sulfate) and neutral proteases (tryptase and/or chymase).	Abbas 11e p.467	*“Histamine ... Enzymes: neutral proteases (tryptase and/or chymase), acid hydrolases, cathepsin G, carboxypeptidase”*		
M2-F073	Mast cells and basophils make the lipid mediators prostaglandin D2 (PGD2), leukotrienes C4, D4, E4 and platelet-activating factor (PAF) on activation.	Abbas 11e p.467	*“Major lipid mediators produced on activation PGD2 ... Leukotrienes C4, D4, E4 ... PAF”*		
M2-F074	Complement fragments C3a and C5a (anaphylatoxins) cause mast cell degranulation; TLR ligands, ATP and cytokines such as SCF and IL-33 activate mast cells to make lipid mediators and cytokines, but not to degranulate.	Abbas 11e p.470	*“C3a and C5a can cause mast cell degranulation ... induce selective activation of mast cells to produce arachidonic acid metabolites, cytokines, and chemokines, but not degranulation”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F075	MRGPRX2 on skin mast cells binds some antibiotics, anaesthetic drugs and venom components, activates mast cells without IgE, and likely mediates some allergy-like drug reactions.	Abbas 11e p.470	*“Ligands that bind to MRGPRX2 and thereby activate mast cells include several antibiotics and anesthetic drugs, components of insect venoms”*		
M2-F076	The inhibitory receptor FcgammaRIIB (ITIM) co-aggregates with FcepsilonRI and inhibits mast cell signalling.	Abbas 11e p.470	*“One such inhibitory receptor is FcγRIIB, which coaggregates with FcεRI during mast cell activation”*		
M2-F077	Tryptase is present in all human mast cells and in no other known cell type, so serum tryptase is a marker of mast cell activation used to diagnose anaphylaxis.	Abbas 11e p.471	*“Tryptase is present in all human mast cells and is not known to be present in any other cell type ... serum tryptase assays have”*		
M2-F078	Human mast cell subsets contain tryptase only or tryptase and chymase.	Abbas 11e p.465	*“granule protease content (tryptase only or tryptase and chymase in humans)”*		
M2-F079	Systemic mastocytosis is a clonal mast cell disorder associated with c-KIT mutations and increased mediator release including tryptase.	Abbas 11e p.470-471	*“Systemic mastocytosis (SM) is a clinically heterogeneous disease associated with mutations in the tyrosine kinase c-KIT”*		
M2-F080	Basophils are blood granulocytes (0.5% or less of leukocytes) that are normally absent from tissues but can be recruited to inflammatory sites.	Abbas 11e p.465	*“Basophils constitute 0.5% or less of blood leukocytes. Although they are normally not present in tissues, basophils may be recruited”*		
M2-F081	Basophils express FcepsilonRI, bind IgE and can be triggered by antigen binding to IgE.	Abbas 11e p.465	*“Like mast cells, basophils express Fcε receptor type I (FcεRI), bind IgE, and can be triggered by antigen binding to the IgE”*		
M2-F082	GM-CSF, IL-3 and IL-5 promote eosinophil maturation from myeloid precursors.	Abbas 11e p.20	*“The cytokines GM-CSF, interleukin-3 (IL-3), and IL-5 promote eosinophil maturation from myeloid precursors”*		
M2-F083	Eosinophils carry Fc receptors for IgA and IgG, TLRs and IL-5 receptors that can trigger granule release.	Abbas 11e p.20	*“including Fc receptors for IgA and IgG, TLRs, and IL-5 receptors, can generate signals that activate the cells to release their granule contents”*		
M2-F084	Eosinophils normally reside in the mucosa of the respiratory, gastrointestinal and genitourinary tracts.	Abbas 11e p.20	*“especially in mucosal linings of the respiratory, gastrointestinal, and genitourinary tracts”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F085	Eosinophil granules contain major basic protein (MBP) and eosinophil cationic protein (ECP), which damage helminths, bacteria and host cells.	Abbas 11e p.473	*“major basic protein and eosinophil cationic protein ... damage the integument of helminths and the cell walls of bacteria, as well as cells in normal”*		
M2-F086	Eosinophil peroxidase is distinct from neutrophil myeloperoxidase and makes hypochlorous or hypobromous acid.	Abbas 11e p.473	*“eosinophil peroxidase, which is distinct from the myeloperoxidase found in neutrophils and catalyzes the production of hypochlorous or hypobromous acid”*		
M2-F087	Eosinophils produce large amounts of galectin-10, which forms Charcot-Leyden crystals at sites of chronic type 2 inflammation such as asthmatic bronchi.	Abbas 11e p.473	*“Eosinophils produce large amounts of the protein galectin-10 ... Charcot-Leyden crystals ... such as the bronchi of allergic asthma patients”*		
M2-F088	Eotaxin (CCL11) binds CCR3 on eosinophils; VCAM-1/VLA-4, C5a, PAF and LTB4 also recruit eosinophils.	Abbas 11e p.473	*“eotaxin (CCL11) ... binds to the chemokine receptor CCR3 expressed on eosinophils”*		
M2-F089	IL-5 from Th2 cells and ILC2s increases eosinophil production in bone marrow and activates mature eosinophils.	Abbas 11e p.473	*“Both Th2 cells and ILC2s are sources of IL-5, which stimulates bone marrow production of eosinophils and is a potent activator of mature eosinophils”*		
M2-F090	Eosinophils express very low FcepsilonRI that lacks a signalling chain; there is no convincing evidence that IgE-antigen activates them.	Abbas 11e p.473	*“eosinophils express very low levels of FcεRI, and the receptor lacks a signaling chain, so its function in these cells is unclear”*		
M2-F091	Neutrophils, Th2 cells and eosinophils are recruited in the late-phase reaction, a few hours to 2 days after allergen exposure.	Abbas 11e p.465	*“late-phase reactions, which occur a few hours to two days after allergen exposure”*		
M2-F092	ILCs secrete cytokines similar to those of CD4+ helper T cells; they are rare in blood and live mainly in mucosal tissues such as lung and intestine.	Abbas 11e p.30	*“ILCs are rare in the blood and are present mostly in tissues, especially mucosal tissues such as the lung and intestines”*		
M2-F093	A common ILC precursor is identified by the transcription factor Id2.	Abbas 11e p.81	*“A common ILC precursor identified by the Id2 transcription factor differentiates into three major subsets”*		
M2-F094	ILC1s make IFN-gamma (T-BET); ILC2s make mainly IL-5 and IL-13 (GATA3); ILC3s make IL-22 and/or IL-17 (RORgammat).	Abbas 11e p.81	*“ILC1s produce IFN-γ and express the transcription factor T-BET ... ILC2s produce mainly IL-5 and IL-13 and express ... GATA3 ... ILC3s produce IL-22 and/or”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F095	ILC subsets are activated by cytokines: ILC1 by IL-12, IL-15, IL-18; ILC2 by IL-25, IL-33, TSLP; ILC3 by IL-1, IL-23.	Abbas 11e p.81	*“each ILC subset is activated by different cytokines (see Fig. 4.8)”*		
M2-F096	ILC1s defend against intracellular microbes; ILC2s against helminths and contribute to allergy; ILC3s defend against extracellular fungi and bacteria and maintain epithelial barriers.	Abbas 11e p.81-82	*“ILC2s may play a role in defense against helminthic parasites and also contribute to allergic diseases”*		
M2-F097	NK cells resemble ILC1s because they secrete IFN-gamma, but ILC1s are not cytotoxic.	Abbas 11e p.82	*“They are developmentally related to ILC1s but are not considered identical because ILC1s do not have cytotoxic functions”*		
M2-F098	Epithelial cells release IL-25, IL-33 and TSLP, which stimulate ILC2s, Th2 cells and mast cells to produce IL-4, IL-5 and IL-13; IL-33 is an alarmin stored in epithelial nuclei.	Abbas 11e p.94	*“IL-25, thymic stromal lymphopoietin (TSLP), and IL-33 ... stimulate ILC2s, Th2 cells, and mast cells to produce IL-4, IL-5, and IL-13”*		
M2-F099	Because ILC2s reside in tissues, they may drive early allergic inflammation before Th2 cells arrive and may later sustain it with Th2 cells.	Abbas 11e p.465	*“their cytokines may contribute to early allergic inflammation before Th2 cells are generated and migrate to the tissues”*		
M2-F100	iNKT cells, gammadelta T cells, MAIT cells, B-1 cells and marginal-zone B cells have antigen receptors of limited diversity and behave more like innate cells.	Abbas 11e p.85-86	*“T cells with limited antigen receptor diversity include invariant natural killer T (iNKT) cells, γδ T cells, mucosa-associated invariant T (MAIT) cells”*		
M2-F101	MAIT cells recognise bacterial riboflavin metabolites and make up almost 50% of liver lymphocytes.	Abbas 11e p.24	*“Limited specificity for bacterial-derived riboflavin metabolites ... In the liver, almost 50% of the lymphocytes are MAIT cells”*		
M2-F102	iNKT cells recognise glycolipids presented by CD1 and rapidly secrete IL-4, IL-10 and IFN-gamma.	Janeway 8e p.119	*“able to recognize glycolipid antigens presented to them by the MHC-like molecule CD1 ... rapid secretion of cytokines, including IL-4, IL-10, and IFN-γ”*		
M2-F103	Intraepithelial gammadelta T cells recognise target molecules directly rather than peptide-MHC complexes.	Janeway 8e p.119	*“γ:δ T cells do not generally recognize antigen as peptides presented by MHC molecules; instead, they seem to recognize their target antigens directly”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F104	B-1 cells make IgM against polysaccharides without T-cell help within 48 hours; in mice they protect against Streptococcus pneumoniae.	Janeway 8e p.119	*“can produce antibodies of the IgM class without ‘help’ from T cells ... they first appear within 48 hours of exposure to antigen”*		
M2-F105	Chronic granulomatous disease (CGD) is caused by defective phagocyte oxidase and production of reactive oxygen species, with recurrent bacterial and fungal infections.	Abbas 11e p.487	*“Chronic granulomatous disease Defective production of reactive oxygen species by phagocytes; recurrent bacterial and fungal infections”*		
M2-F106	Leukocyte adhesion deficiency type 1 (LAD-1) is caused by mutations in CD18, the beta chain of beta2 integrins.	Abbas 11e p.487	*“Mutations in gene encoding the β chain (CD18) of β2 integrins”*		
M2-F107	Mast cells, basophils and eosinophils are myeloid cells with granules full of inflammatory mediators; all three take part in defence against helminths and in allergic disease.	Abbas 11e p.20	*“All three share the common property of having cytoplasmic granules filled with various inflammatory and antimicrobial mediators”*		
M2-F108	FcepsilonRII (CD23) is a C-type lectin-like IgE receptor with much lower affinity than FcepsilonRI; its biologic role is not known.	Abbas 11e p.467	*“Another IgE receptor called FcεRII, also known as CD23, is a protein related to C-type mammalian lectins whose affinity for IgE is much lower”*		
M2-F109	In Abbas Table 20.1, mast cell granules contain histamine, heparin and/or chondroitin sulfate and protease; basophil granules contain histamine, chondroitin sulfate and protease.	Abbas 11e p.466	*“Histamine, heparin, and/ or chondroitin sulfate, protease ... Histamine, chondroitin sulfate, protease”*		
M2-F110	Activated eosinophils make lipid mediators, including PAF, prostaglandins and cysteinyl leukotrienes, and cytokines such as IL-4 and TGF-beta.	Abbas 11e p.473	*“Activated eosinophils, like mast cells and basophils, produce and release lipid mediators, including PAF, prostaglandins, and the cysteinyl leukotrienes”*		
M2-F111	In Abbas Table 20.2, galectin-10 is an eosinophil mediator stored preformed in the cytosol.	Abbas 11e p.467	*“Preformed in cytosol ... Galectin-10 ... Formation of Charcot-Leyden crystals”*		
M2-F112	Abbas Chapter 4 states that TLR ligands can induce mast cell degranulation (this differs from Abbas p.470).	Abbas 11e p.86	*“Mast cells express TLRs and TLR ligands can induce mast cell degranulation”*		
M2-F113	ILCs are bone marrow-derived cells with lymphocyte morphology that lack T-cell receptors (TCRs).	Abbas 11e p.81	*“bone marrow–derived cells with lymphocyte morphology that were discovered as cells that produced cytokines similar to those made by helper T cells but lacked TCRs”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M2-F114	pDCs develop in the bone marrow from a precursor distinct from that of conventional DCs; some pDCs may arise from the common DC precursor.	Abbas 11e p.21-22	*“They develop in the bone marrow from a precursor distinct from that for conventional DCs ... Some plasmacytoid DCs may arise from the common DC precursor”*		
M3-F001	Microbial molecules that stimulate innate immunity and are shared by classes of microbes are called pathogen-associated molecular patterns (PAMPs).	Abbas 11e p.63	*“The microbial substances that stimulate innate immunity are often shared by classes of microbes and are called pathogen-associated molecular patterns (PAMPs).”*		
M3-F002	Typical PAMPs are double-stranded RNA of replicating viruses, unmethylated CpG DNA of bacteria, N-formylmethionine-initiated proteins, LPS, lipoteichoic acid and terminal-mannose oligosaccharides.	Abbas 11e p.63	*“long double-stranded RNA found in replicating viruses and unmethylated CpG DNA sequences found in bacteria”*		
M3-F003	Damage-associated molecular patterns (DAMPs) are endogenous molecules that damaged or dying cells release; they can signal sterile injury such as burns, trauma or ischaemia.	Abbas 11e p.63	*“These substances are called damage-associated molecular patterns (DAMPs)”*		
M3-F004	Examples of DAMPs are heat shock proteins, monosodium urate crystals, extracellular formylated peptides and ATP, extracellular HMGB1, histones and cytoplasmic dsDNA.	Abbas 11e p.65	*“Extracellular HMGB1, histones, cytoplasmic dsDNA”*		
M3-F005	Cellular pattern recognition receptors (PRRs) are located on the cell surface, in phagocytic vesicles (endosomes) and in the cytosol; soluble recognition molecules are in blood and secretions.	Abbas 11e p.64	*“These receptors are located on the surfaces, in phagocytic vesicles, and in the cytosol of various cell types”*		
M3-F006	Humans have 10 functional TLRs, TLR1 to TLR10.	Abbas 11e p.65	*“There are 10 different functional TLRs in humans, named TLR1 through TLR10”*		
M3-F007	The function of TLR10 is poorly understood; it may have anti-inflammatory functions, unlike all other TLRs.	Abbas 11e p.67	*“there is some evidence that it may have anti-inflammatory functions, unlike all the other TLRs”*		
M3-F008	The Toll/IL-1 receptor (TIR) domain in the TLR cytoplasmic tail is also present in the receptors for IL-1, IL-18 and IL-33.	Abbas 11e p.67	*“TIR domains are also found in the cytoplasmic tails of the receptors for the cytokines IL-1, IL-18, and IL-33”*		
M3-F009	TLR1, TLR2, TLR4, TLR5 and TLR6 are on the plasma membrane and recognize extracellular bacterial and fungal PAMPs.	Abbas 11e p.69	*“TLRs 1, 2, 4, 5, and 6 are expressed on the plasma membrane”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F010	TLR3, TLR7, TLR8 and TLR9 are mainly on endosomal membranes and detect microbial nucleic acids.	Abbas 11e p.69	*“TLRs 3, 7, 8, and 9 are mainly expressed inside cells on endosomal membranes”*		
M3-F011	Endosomal TLRs distinguish microbial from host nucleic acids by location, because host RNA is not normally present in endosomes.	Abbas 11e p.69	*“the endosomal TLRs may distinguish nucleic acids of normal cells from microbial nucleic acids on the basis of the cellular location”*		
M3-F012	LPS of gram-negative bacteria binds TLR4; peptidoglycan and lipoteichoic acid of gram-positive bacteria bind TLR2.	Abbas 11e p.67	*“LPS of gram-negative bacteria, which binds TLR4; and peptidoglycan and lipoteichoic acid of gram-positive bacteria, which bind TLR2.”*		
M3-F013	Flagellin of motile bacteria binds TLR5.	Abbas 11e p.67	*“flagellin, a protein subunit component of the flagella of motile bacteria, which binds to TLR5.”*		
M3-F014	Double-stranded RNA binds TLR3; viral single-stranded RNA in endosomes binds TLR7 and TLR8.	Abbas 11e p.67	*“double-stranded RNAs, which are present in viruses but not mammalian cells, bind to TLR3”*		
M3-F015	Unmethylated CpG motifs, common in prokaryotic DNA but rare in vertebrate genomes, bind TLR9.	Abbas 11e p.67	*“unmethylated CpG nucleotide repeats, which are common in prokaryotic DNA but rare in vertebrate genomes, bind to TLR9.”*		
M3-F016	In mammalian genomic DNA, CpG dinucleotides are heavily methylated; in bacteria and many viruses they stay unmethylated and act as a PAMP.	Janeway 8e p.88	*“In mammalian genomes, CpG dinucleotides in genomic DNA are heavily methylated on the cytosine by DNA methyltransferases.”*		
M3-F017	TLR9 recognizes unmethylated CpG DNA of bacteria and herpesviruses and is expressed by plasmacytoid dendritic cells, B cells, eosinophils and basophils.	Janeway 8e p.86	*“TLR-9 DNA with unmethylated CpG (bacteria and herpesviruses) Plasmacytoid dendritic cells, eosinophils, B cells, basophils”*		
M3-F018	TLR9-deficient plasmacytoid dendritic cells cannot make type I interferons in response to herpes simplex virus.	Janeway 8e p.112	*“the inability of TLR-9-deficient plasmacytoid dendritic cells to generate type I interferons in response to herpes simplex virus”*		
M3-F019	The TLR1:TLR2 heterodimer recognizes triacyl lipopeptides and the TLR2:TLR6 heterodimer recognizes diacyl lipopeptides.	Janeway 8e p.87	*“The diacyl and triacyl lipopeptides recognized by the heterodimeric receptors TLR-6:TLR-2 and TLR-1:TLR-2, respectively”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F020	LPS first binds soluble LPS-binding protein; the accessory protein MD2 binds lipid A of LPS and the complex then engages TLR4.	Abbas 11e p.69	*“An extracellular protein called MD2 (myeloid differentiation protein 2) binds to the lipid A component of LPS”*		
M3-F021	CD14 is also required for efficient LPS-induced TLR4 signalling.	Abbas 11e p.69	*“Another protein, called CD14, is also required for efficient LPS-induced signaling.”*		
M3-F022	The protein UNC93B moves TLRs from the endoplasmic reticulum to endosomes; deficiency of TLR3 or UNC93B causes susceptibility to herpes simplex virus encephalitis.	Abbas 11e p.69	*“Genetic deficiency in TLR3 or UNC93B leads to susceptibility to certain viral infections, especially herpes simplex virus encephalitis”*		
M3-F023	NF-κB induces inflammatory cytokines (TNF, IL-1), chemokines (CCL2, CXCL8) and endothelial adhesion molecules (E-selectin).	Abbas 11e p.69	*“inflammatory cytokines (e.g., tumor necrosis factor [TNF] and IL-1), chemokines (e.g., CCL2 and CXCL8), and endothelial adhesion molecules (e.g., E-selectin)”*		
M3-F024	IRF3 and IRF7, together with NF-κB, induce the type I interferons IFN-α and IFN-β.	Abbas 11e p.69	*“IRF3 and IRF7, together with NF-κB, promote the expression of genes encoding interferon (IFN)-α and IFN-β”*		
M3-F025	All TLRs except TLR3 use the adaptor MyD88; only TLR3 and TLR4 use the adaptor TRIF.	Abbas 11e p.69	*“All TLRs except TLR3 engage the adaptor MyD88, while only TLR3 and TLR4 utilize the adaptor TRIF”*		
M3-F026	Mammalian TLRs use four TIR-containing adaptors: MyD88, MAL, TRIF and TRAM.	Janeway 8e p.90	*“There are four such adaptors used by mammalian TLRs: MyD88 (myeloid differentiation factor 88), MAL”*		
M3-F027	TLR5, TLR7 and TLR9 use only MyD88; TLR3 uses only TRIF; TLR2 heterodimers use MyD88/MAL; TLR4 uses MyD88/MAL and TRIF/TRAM.	Janeway 8e p.90	*“TLR-4 signaling uses both of these adaptor pairs, MyD88/MAL and TRIF/TRAM.”*		
M3-F028	MyD88 binds the TLR through its TIR domain; the MyD88 death domain recruits and activates the kinases IRAK4 and IRAK1.	Janeway 8e p.91	*“the death domain recruits and activates two serine-threonine protein kinases—IRAK4 (IL1-receptor associated kinase 4) and IRAK1”*		
M3-F029	IRAK recruits TRAF6, which leads to activation of TAK1 and the IKK complex (IKKα, IKKβ, IKKγ/NEMO); IKK phosphorylates IκB, which releases NF-κB to induce TNF, IL-1β and IL-6.	Janeway 8e p.91	*“Activated IKK phosphory-lates IkB, which then releases NF«B.”*		
M3-F030	The endosomal TLR7, TLR8 and TLR9 also use MyD88, but this signalling activates IRFs and induces type I IFN antiviral responses.	Abbas 11e p.69	*“The endosomal TLRs 7, 8, and 9 also engage MyD88”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F031	MyD88 binds IRF7 directly in endosomes, so TLR7 and TLR9 induce IFN-α and IFN-β in plasmacytoid dendritic cells.	Janeway 8e p.92	*“the endosomal TLRs—TLR-7 and TLR-9—can activate IRF7, resulting in the production of IFN-a and IFN-f by cells such as plasmacytoid dendritic cells”*		
M3-F032	Cell-surface TLR4 uses both MyD88 and TRIF and activates NF-κB, IRF3 and IRF7, giving inflammatory and antiviral responses.	Abbas 11e p.69	*“Cell surface TLR4 engages both MyD88 and TRIF, leading to NF-κB, IRF3 and IRF7 activation”*		
M3-F033	Endosomal TLR3 signals only through TRIF, which activates NF-κB and IRF3.	Abbas 11e p.69	*“Endosomal TLR3 signals only through TRIF, which activates NF-κB and IRF3.”*		
M3-F034	TRIF activates the kinases IKKε and TBK1, which phosphorylate IRF3 and induce IFN-β.	Janeway 8e p.92	*“binds and activates the kinases IkKe and TBK1. These in turn phosphorylate IRF3, which induces the expression of IFN-B.”*		
M3-F035	TLR activation induces the co-stimulatory molecules CD80 (B7.1) and CD86 (B7.2) on dendritic cells and macrophages, which are essential to start adaptive immunity.	Janeway 8e p.96	*“called B7.1 (CD80) and B7.2 (CD86), are expressed by both macrophages and tissue dendritic cells in response to activation by pathogen sensors such as TLRs”*		
M3-F036	Patients with MyD88 or IRAK4 mutations have severe invasive bacterial infections early in life, especially pneumococcal pneumonia; infections become less severe later.	Abbas 11e p.489	*“Individuals with mutations in MyD88 and IRAK4 have severe, invasive bacterial infections early in life, especially pneumococcal pneumonia.”*		
M3-F037	Patients with MyD88 or IRAK4 deficiency usually cannot mount a fever during illness.	Abbas 11e p.489	*“these patients are usually unable to mount a fever in the setting of illness.”*		
M3-F038	Autosomal recessive MyD88 or IRAK4 deficiency causes recurrent invasive pyogenic infections with little inflammation ('cold' infections).	Janeway 8e p.535	*“recurrent, severe peripheral, and invasive infections by pyogenic bacteria that elicit little inflammation, a situation known as a ‘cold’ infection”*		
M3-F039	In human IRAK4 deficiency, no extra susceptibility to viral infections is seen.	Janeway 8e p.92	*“In human IRAK4 deficiency, for example, no extra susceptibility to viral infections has been noted.”*		
M3-F040	Loss-of-function TLR3 mutations cause recurrent herpes simplex encephalitis without a general increase in other viral infections.	Abbas 11e p.489	*“TLR3 loss of function mutations can result in recurrent herpes simplex encephalitis.”*		
M3-F041	Autosomal recessive TRIF mutations and autosomal dominant TRAF3 or TBK1 mutations also cause susceptibility to herpes simplex encephalitis.	Abbas 11e p.489	*“Autosomal recessive mutations in TRIF and autosomal dominant mutations affecting the TRAF3 E3 ligase also result*		

ID	Fact	Source	Supporting quote	Vol.	Verified
			in susceptibility to herpes simplex encephalitis."		
M3-F042	Homozygous UNC93B mutations reduce type I IFN production and increase susceptibility to herpes simplex encephalitis.	Abbas 11e p.489	*"Homozygous mutations in UNC93B result in reduced type 1 IFN generation and increased susceptibility to herpes simplex encephalitis."*		
M3-F043	NEMO deficiency blocks NF-κB activation downstream of all TLRs and so affects both innate and adaptive immunity.	Janeway 8e p.535	*"also impairs TLR signaling though its block of normal NF«B activation downstream of all TLRs"*		
M3-F044	X-linked EDA-ID (anhidrotic ectodermal dysplasia with immunodeficiency), usually from NEMO or IKKβ mutations, impairs TLR and CD40 signalling and causes infections with pyogenic bacteria, mycobacteria, viruses and Pneumocystis.	Abbas 11e p.498	*"These patients suffer from infections with encapsulated pyogenic bacteria as well as with intracellular bacterial pathogens, including mycobacteria, viruses, and fungi such as P. jirovecii."*		
M3-F045	NOD1 recognizes diaminopimelic acid (DAP), mainly from gram-negative peptidoglycan.	Abbas 11e p.72	*"NOD1 recognizes a glycosylated tripeptide called diaminopimelic acid (DAP), derived mainly from gram-negative bacterial peptidoglycan"*		
M3-F046	NOD2 recognizes muramyl dipeptide (MDP) from peptidoglycan of gram-negative and gram-positive bacteria.	Abbas 11e p.72	*"NOD2 recognizes a distinct molecule called muramyl dipeptide (MDP), derived from peptidoglycans of both gram-negative and gram-positive bacteria."*		
M3-F047	Activated NOD1/NOD2 recruit the kinase RIP2, which activates NF-κB.	Abbas 11e p.72	*"the CARD effector domains of the NOD proteins to recruit many molecules of the kinase RIP2"*		
M3-F048	Some NOD2 polymorphisms increase the risk of Crohn's disease, probably through defective sensing of intestinal microbes.	Abbas 11e p.72	*"Certain NOD2 gene polymorphisms increase the risk for an inflammatory disease of the bowel called Crohn's disease."*		
M3-F049	Gain-of-function NOD2 mutations cause the systemic inflammatory disease Blau syndrome.	Abbas 11e p.72	*"gain-of-function mutations of NOD2 that cause increased NOD signaling and NF-κB activation lead to a systemic inflammatory disease called Blau syndrome"*		
M3-F050	Blau syndrome causes spontaneous inflammation in the joints, eyes and skin.	Janeway 8e p.93	*"Blau syndrome, which is characterized by spontaneous inflammation in the joints, eyes, and skin."*		
M3-F051	Inflammasomes are cytosolic multiprotein complexes that form in response to infection or cell injury and generate active caspase-1.	Abbas 11e p.72	*"Inflammasomes are multiprotein enzymatic complexes that form in the cytosol in response to infections or cell injury"*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F052	Oligomerized NLRP3 binds the adaptor ASC; ASC filaments recruit procaspase-1, which clusters and becomes active caspase-1.	Abbas 11e p.72	*“each NLRP3 protein in the oligomer binds an adaptor protein called ASC (apoptosis-associated speck-like protein containing a CARD).”*		
M3-F053	The main function of caspase-1 is to cleave the inactive precursors of IL-1β and IL-18 into active cytokines.	Abbas 11e p.72	*“the main function of caspase-1 is to cleave the inactive cytoplasmic precursor forms of IL-1β and IL-18.”*		
M3-F054	IL-1β lacks a signal peptide; caspase-1 cleaves gasdermin D, and the N-terminal fragment forms plasma membrane pores through which IL-1β leaves the cell.	Abbas 11e p.72	*“caspase-1 cleavage of a cytosolic protein called gasdermin D, removing the inhibitory C-terminal domain and generating an N-terminal fragment that polymerizes to form pores”*		
M3-F055	IL-1β production needs two signals: signal 1 (TLR, NLR or RLR signalling through NF-κB) makes pro-IL-1β; signal 2 activates the inflammasome to cleave it.	Abbas 11e p.93	*“IL-1 production usually requires two distinct signals: one that activates new gene transcription and production of a 33-kD precursor pro–IL-1β polypeptide”*		
M3-F056	NLRP3 senses uric acid crystals, alum, cholesterol crystals, extracellular ATP, silica, streptococcal and staphylococcal toxins, and influenza virus.	Abbas 11e p.74	*“uric acid crystals, aluminum hydroxide crystals used in vaccine adjuvants, cholesterol crystals, adenosine triphosphate (ATP) released from mitochondria, silica”*		
M3-F057	Common upstream triggers of NLRP3 are reduced cytosolic K+ (K+ efflux) and reactive oxygen species; crystals may act by damaging lysosomal membranes.	Abbas 11e p.74	*“Reduced cytoplasmic potassium ion concentration may be one such common mechanism”*		
M3-F058	The adjuvant effect of aluminium hydroxide (alum) depends on NLRP3 and the inflammasome pathway.	Janeway 8e p.94	*“aluminum hydroxide (alum) rely on NALP3 and the inflammasome pathway”*		
M3-F059	Pyroptosis is an inflammatory programmed cell death of macrophages and dendritic cells, but not neutrophils, caused by gasdermin D pores; fluid enters, the cell swells and the plasma membrane fails.	Abbas 11e p.74	*“an inflammatory form of programmed cell death of macrophages and DCs (but not of neutrophils and most other cell types) called pyroptosis”*		
M3-F060	In the noncanonical pathway, cytosolic LPS activates caspase-4 or caspase-5 in humans (caspase-11 in rodents), which cleaves gasdermin D and causes pyroptosis.	Abbas 11e p.74	*“uses a different caspase (caspase-11 in rodents; caspase-4 or caspase-5 in humans)”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F061	Other inflammasome sensors are NLRC4 (cytosolic flagellin, type III secretion system), NLRP1 (anthrax lethal toxin), AIM2 (dsDNA) and pyrin (toxins that modify Rho GTPases).	Abbas 11e p.72-74	*“NLRC4 recognizes cytosolic flagellin and components of the type III secretion system of bacteria. NLRP1 recognizes the anthrax lethal toxin.”*		
M3-F062	Phagocytosed monosodium urate crystals damage lysosomes and activate inflammasomes; IL-1 antagonists are used for severe gout resistant to conventional drugs.	Abbas 11e p.74	*“IL-1 antagonists have been used to effectively treat cases of severe gout that are resistant to conventional anti-inflammatory drugs.”*	⚠	
M3-F063	Gain-of-function mutations in inflammasome components, or loss-of-function mutations in their regulators, cause excess IL-1 with recurrent fever and inflammation of skin, joints and abdomen (inflammasomopathies, a subgroup of autoinflammatory syndromes).	Abbas 11e p.74	*“due to autosomal gain-of-function mutations in one or another of their component proteins or loss-of-function mutations in their regulators”*		
M3-F064	Autoinflammatory diseases differ from autoimmune diseases, which are caused by self-reactive antibodies and/or T cells.	Abbas 11e p.74	*“Autoinflammatory diseases are distinct from autoimmune diseases, which are disorders of adaptive immunity caused by antibodies and/or T cells reactive with self antigens.”*		
M3-F065	Mutations in NLRP3 (cryopyrin) cause cryopyrin-associated periodic syndromes (CAPS).	Abbas 11e p.74	*“Autoinflammatory diseases caused by mutations in NLRP3 (also known as cryopyrin) are called cryopyrin-associated periodic syndromes (CAPS).”*		
M3-F066	Patients with CAPS can be treated successfully with IL-1 antagonists.	Abbas 11e p.74	*“Patients with CAPS can be successfully treated with IL-1 antagonists”*	⚠	
M3-F067	CAPS is caused by gain-of-function NLRP3 mutations and presents with fever, neutrophilic rash, bony lesions and central nervous system inflammation.	Abbas 11e p.75	*“Fever, neutrophilic rash, bony lesions, central nervous system inflammation”*		
M3-F068	Muckle-Wells syndrome and familial cold autoinflammatory syndrome (FCAS) are dominantly inherited NLRP3 disorders with fever (cold-induced in FCAS), urticarial rash, joint pain and conjunctivitis.	Janeway 8e p.537	*“episodes of fever—which is induced by exposure to cold in the case of FCAS—as well as urticarial rash, joint pains, and conjunctivitis”*		
M3-F069	Muckle-Wells syndrome also causes progressive deafness.	Janeway 8e p.537	*“Periodic fever, urticarial rash, joint pains, conjunctivitis, progressive deafness”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F070	NLRP3 mutations also cause chronic infantile neurologic cutaneous and articular syndrome (CINCA), with neonatal-onset fever, urticarial rash, arthropathy and neurological disease.	Janeway 8e p.537	*“Mutations in NLRP3 are also associated with the autoinflammatory disorder chronic infantile neurologic cutaneous and articular syndrome (CINCA)”*		
M3-F071	Muckle-Wells syndrome responds dramatically to anakinra, an IL-1 receptor antagonist.	Janeway 8e p.537	*“Muckle–Wells syndrome responds dramatically to the drug anakinra, an antagonist of the receptor for IL-1.”*	⚠	
M3-F072	Familial Mediterranean fever (FMF) is caused by mutations in MEFV, which encodes pyrin; it presents with fever, serositis and arthritis.	Abbas 11e p.74-75	*“familial Mediterranean fever, caused by mutation of the MEFV gene, which encodes pyrin.”*		
M3-F073	TNF receptor-associated periodic syndrome (TRAPS) is caused by mutations in the gene for TNF receptor I, gives periodic fever, myalgia and rash, and responds to TNF blockade.	Janeway 8e p.536-537	*“(TRAPS) is caused by mutations in quite a different gene, that encoding the TNF-a receptor TNFR-I.”*	⚠	
M3-F074	Hyper-IgD syndrome (HIDS) is caused by partial mevalonate kinase deficiency and gives fever from infancy, high serum IgD and lymphadenopathy.	Janeway 8e p.537	*“is caused by mutations that result in a partial deficiency of mevalonate kinase”*		
M3-F075	RIG-like receptors (RLRs) are cytosolic sensors of viral RNA that induce type I IFNs; the best known are RIG-I and MDA5.	Abbas 11e p.75	*“RIG-like receptors (RLRs) are cytosolic sensors of viral RNA that respond by inducing the production of the antiviral type I IFNs”*		
M3-F076	MDA5 recognizes long dsRNA (1-6 kb).	Abbas 11e p.75	*“MDA5 recognizes long dsRNA (1–6 kb)”*		
M3-F077	RIG-I recognizes RNA with a 5′-triphosphate, which host cytosolic mRNA does not have because of its 7-methylguanosine cap.	Abbas 11e p.75-76	*“RIG-I will only recognize RNA with a 5′ triphosphate moiety”*		
M3-F078	SARS-CoV-2 RNA vaccines use pseudouridine instead of uridine to reduce detection by innate RNA sensors.	Abbas 11e p.76	*“changing uridines to derivatives called pseudouridines, which reduced detection by innate RNA sensors”*		
M3-F079	Activated RLRs bind MAVS on the outer mitochondrial membrane; MAVS polymerizes and recruits TRAF proteins that activate TBK1 and IKK.	Abbas 11e p.76	*“the RLRs are recruited to the outer mitochondrial membrane by the MAVS (mitochondrial antiviral-signaling) protein”*		
M3-F080	TBK1 phosphorylates IRF3 and IRF7, and IKK activates NF-κB; these transcription factors induce type I IFNs.	Abbas 11e p.76	*“TBK1 phosphorylates and activates IRF3 and IRF7 and IKK activates NF-κB; these transcription factors induce*		

ID	Fact	Source	Supporting quote	Vol.	Verified
			production of type I IFNs."		
M3-F081	RIG-I-deficient mice are susceptible to several ssRNA viruses but not picornaviruses; MDA5-deficient mice are susceptible to picornaviruses.	Janeway 8e p.95	*"mice deficient in MDA-5 are susceptible to picornaviruses, indicating that these two sensors of viral RNAs have crucial but distinct roles"*		
M3-F082	Cytosolic DNA sensors detect dsDNA from microbes or damaged host cells (micronuclei, mitochondria) and induce type I IFN and autophagy.	Abbas 11e p.76	*"Cytosolic DNA sensors (CDSs) are molecules that detect double-stranded (ds) DNA in the cytosol"*		
M3-F083	Cytosolic dsDNA activates the enzyme cGAS, which makes the second messenger cyclic GMP-AMP (cGAMP).	Abbas 11e p.76	*"cytosolic dsDNA activates the enzyme cGAS (cyclic guanosine monophosphate–adenosine monophosphate [GMP-AMP] synthase), which generates a signaling molecule called cGAMP"*		
M3-F084	STING is an endoplasmic reticulum adaptor that binds cGAMP, moves to the Golgi and activates TBK1, which activates IRF3/IRF7 and NF-κB to induce type I IFN.	Abbas 11e p.76	*"STING is an endoplasmic reticulum–localized transmembrane adaptor protein whose cytosolic tail binds to cGAMP with high affinity"*		
M3-F085	Gain-of-function STING mutations cause a type I IFN-driven systemic inflammatory disease of skin and lung (an interferonopathy).	Abbas 11e p.77	*"Gain-of-function mutations in the gene that encodes STING result in systemic inflammatory disease with manifestations in skin and lung"*		
M3-F086	STING-associated vasculopathy with onset in infancy (SAVI) presents with cutaneous vasculopathy, fever and interstitial lung disease.	Abbas 11e p.75	*"Cutaneous vasculopathy, fever, and interstitial lung disease"*		
M3-F087	Aicardi-Goutières syndrome is an interferonopathy caused by loss-of-function mutations in nucleases (e.g., TREX1, SAMHD1), so excess nucleic acid reaches the sensors.	Abbas 11e p.75	*"Loss of function mutations in genes encoding DNAases and RNAases, excessive DNA and RNA available to nucleic acid sensors"*		
M3-F088	AIM2 binds cytosolic dsDNA and forms an inflammasome that generates active IL-1β.	Abbas 11e p.78	*"AIM2 (absent in melanoma-2) is another cytosolic sensor that binds cytosolic dsDNA and forms an inflammasome"*		
M3-F089	The mannose receptor (CD206) on macrophages mediates phagocytosis of microbes with terminal mannose, fucose or N-acetylglucosamine; host glycans usually end in galactose and sialic acid.	Abbas 11e p.78	*"eukaryotic cell surface carbohydrates are most often terminated by galactose and sialic acid"*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F090	Dectin-1 (CD369) binds fungal β-glucan and is required for immunity to Candida, Aspergillus and Pneumocystis.	Abbas 11e p.78	*“Dectin-1 (CD369) binds β-glucan, which is a major cell wall component of many fungal species”*		
M3-F091	Dectin-1 and Dectin-2 signal through an ITAM/SYK/CARD9 pathway to NF-κB and promote Th17 differentiation and antifungal immunity.	Abbas 11e p.78	*“ITAM/SYK/CARD9 pathway of NF-κB activation”*		
M3-F092	Scavenger receptors such as SR-A and CD36 on macrophages mediate phagocytosis; CD36 is a co-receptor for TLR2/6.	Abbas 11e p.79	*“CD36 functions as a coreceptor in TLR2/6 recognition and response to bacterially derived lipoteichoic acid and diacylated lipopeptides”*		
M3-F093	Formyl peptide receptor 1 (FPR1) on leukocytes recognizes N-formylmethionyl peptides, which bacteria and mitochondria make, and directs cell movement.	Abbas 11e p.79	*“synthesis of all bacterial proteins and few mammalian proteins (only those synthesized within mitochondria) is initiated by N-formylmethionine”*		
M3-F094	Pentraxins are the short pentraxins CRP and serum amyloid P (SAP) and the long pentraxin PTX3.	Abbas 11e p.87	*“the short pentraxins, C-reactive protein (CRP) and serum amyloid P (SAP), and the long pentraxin PTX3”*		
M3-F095	CRP binds phosphorylcholine and SAP binds phosphatidylethanolamine on bacteria and apoptotic cells.	Abbas 11e p.87	*“The ligands recognized by CRP and SAP include phosphorylcholine and phosphatidylethanolamine, respectively”*		
M3-F096	CRP, SAP and PTX3 all activate the classical complement pathway by binding C1q.	Abbas 11e p.88	*“CRP, SAP, and PTX3 all activate the classical pathway of complement by binding C1q.”*		
M3-F097	Plasma CRP can rise up to 1000-fold in infection because IL-6, IL-1 and TNF increase its synthesis in the liver.	Abbas 11e p.88	*“Plasma concentrations of CRP are very low in healthy individuals but can increase up to 1000-fold during infections”*		
M3-F098	PTX3 is made by dendritic cells, macrophages and endothelial cells, is stored in neutrophil granules and protects mice against Aspergillus fumigatus and influenza.	Abbas 11e p.88	*“PTX3 is also stored in neutrophil granules and released as neutrophils die.”*		
M3-F099	Mannose-binding lectin (MBL) is a collectin that binds terminal mannose and fucose and acts as an opsonin; low MBL levels increase infection risk, mainly with other immune defects.	Abbas 11e p.88	*“Low MBL levels result in increased susceptibility to a variety of infections, especially in individuals who have other immune defects.”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F100	MBL triggers the lectin pathway of complement through MASP1 and MASP2.	Abbas 11e p.86	*“The lectin pathway is triggered by a plasma protein called mannose-binding lectin (MBL)”*		
M3-F101	The collectins SP-A and SP-D in lung alveoli bind microbes and act as opsonins for alveolar macrophages.	Abbas 11e p.88	*“They bind to various microorganisms and act as opsonins, facilitating ingestion of the microbes by alveolar macrophages.”*		
M3-F102	Ficolins have a fibrinogen-type (not C-type lectin) recognition domain, bind N-acetylglucosamine and lipoteichoic acid, opsonize bacteria and activate complement like MBL.	Abbas 11e p.89	*“instead of a C-type lectin domain, they have a fibrinogen-type carbohydrate recognition domain”*		
M3-F103	TNF, mainly from macrophages, activates endothelium and neutrophils, induces acute-phase proteins and fever, and causes cachexia and apoptosis.	Abbas 11e p.90	*“Muscle, fat: catabolism (cachexia)”*		
M3-F104	IL-1 activates endothelium, causes fever, induces acute-phase proteins and promotes Th17 differentiation.	Abbas 11e p.90	*“Endothelial cells: activation (inflammation, coagulation) Hypothalamus: fever Liver: synthesis of acute-phase proteins T cells: Th17 differentiation”*		
M3-F105	IL-6 induces acute-phase reactants in the liver and promotes differentiation of IL-17-producing helper T cells.	Abbas 11e p.93	*“It induces the synthesis of acute-phase reactants by the liver and promotes the differentiation of IL-17–producing helper T cells.”*		
M3-F106	IL-12 from dendritic cells and macrophages stimulates IFN-γ production by ILC1s, NK cells and T cells and promotes Th1 differentiation.	Abbas 11e p.93	*“it stimulates IFN-γ production by ILC1s, NK cells, and T cells; enhances NK cell–and CTL-mediated cytotoxicity; and promotes differentiation of Th1 cells”*		
M3-F107	Patients with IL-12 receptor β1 mutations are highly susceptible to Salmonella and mycobacteria.	Abbas 11e p.93	*“patients with mutations in the IL-12 receptor β1 subunit are highly susceptible to infections with intracellular bacteria, notably Salmonella and mycobacteria”*		
M3-F108	IL-18, like IL-1, needs the inflammasome for activation; it enhances NK cell functions.	Abbas 11e p.94	*“IL-18 enhances the functions of NK cells, similar to IL-12.”*		
M3-F109	Type I IFNs induce an antiviral state and increase MHC class I expression in all cells, and activate NK cells.	Abbas 11e p.90	*“All cells: antiviral state, increased MHC-I expression”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F110	Plasmacytoid dendritic cells are the major source of IFN-α; many cell types make IFN-β.	Abbas 11e p.98	*“Plasmacytoid DCs are the major sources of IFN-α”*		
M3-F111	TNF and IL-1 stimulate secretion of the chemokines CXCL8 and CCL2, which attract neutrophils and monocytes respectively.	Abbas 11e p.94	*“TNF and IL-1 also stimulate various cells to secrete chemokines, such as CXCL8 and CCL2, which bind to receptors on neutrophils and monocytes, respectively.”*		
M3-F112	TNF, IL-1 and IL-6 induce hepatocytes to produce acute-phase proteins, including CRP, SAP and fibrinogen.	Abbas 11e p.97	*“TNF, IL-1, and IL-6 induce hepatocytes to produce acute-phase proteins, including CRP, SAP, and fibrinogen”*		
M3-F113	Septic shock is mediated in large part by TNF.	Abbas 11e p.92	*“Septic shock, a life-threatening condition resulting from severe infections, is mediated in large part by TNF.”*		
M3-F114	IL-33 is stored in the nuclei of barrier epithelial cells and acts as an alarmin when damaged cells release it.	Abbas 11e p.94	*“It is often called an alarmin because it is rapidly released from damaged epithelial cells”*		
M3-F115	Mast cells express TLRs, and TLR ligands can induce mast cell degranulation.	Abbas 11e p.86	*“Mast cells express TLRs and TLR ligands can induce mast cell degranulation.”*		
M3-F116	In the allergy chapter, TLR ligands are listed among stimuli that make mast cells produce lipid mediators, cytokines and chemokines without degranulation.	Abbas 11e p.470	*“These stimuli include Toll-like receptor (TLR) ligands; substances released from injured cells, including ATP; fungal glucans”*		
M3-F117	Some DAMPs act through TLRs: heat shock proteins bind TLR4 and HMGB1 binds TLR2.	Abbas 11e p.67	*“heat shock proteins (HSPs), which are chaperones induced in response to various cell stresses and bind to TLR4”*		
M3-F118	Type I IFNs from virus-exposed cells act mainly on uninfected neighbouring cells to prevent spread of the virus.	Abbas 11e p.97	*“act mainly on other uninfected cells to prevent the spread of viral infection”*		
M3-F119	The MyD88–IRAK4–IRAK1 pathway lies downstream of most TLRs and of the IL-1 receptor and induces proinflammatory cytokines through NF-κB.	Abbas 11e p.489	*“The major signaling pathway downstream of most TLRs and of the IL-1 receptor (IL-1R) includes the MyD88 adaptor and the IRAK-4 and IRAK-1 kinases”*		
M3-F120	Janeway describes FMF as autosomal recessive and proposes that absence of functional pyrin leads to unregulated cytokine activity.	Janeway 8e p.536-537	*“It is proposed that an absence of functional pyrin leads to unregulated cytokine activity and defective apoptosis”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M3-F121	STING directly senses the bacterial second messengers cyclic di-GMP and cyclic di-AMP.	Abbas 11e p.77	*“The bacterial second messenger molecules cyclic di-GMP (c-di-GMP) and cyclic di-AMP (c-di-AMP) are directly sensed by STING.”*		
M4-F001	Selectins mediate the first, low-affinity adhesion of circulating leukocytes to endothelial cells of postcapillary venules.	Abbas 11e p.44	*“Selectins are plasma membrane carbohydrate-binding adhesion molecules that mediate an initial step of low-affinity adhesion of circulating leukocytes”*		
M4-F002	Endothelial cells express P-selectin (CD62P) and E-selectin (CD62E).	Abbas 11e p.45	*“Endothelial cells express two types of selectins, called P-selectin (CD62P) and E-selectin (CD62E).”*		
M4-F003	Leukocytes, not endothelial cells, express L-selectin (CD62L).	Abbas 11e p.45	*“A third selectin, called L-selectin (CD62L), is expressed on leukocytes and not on endothelial cells.”*		
M4-F004	P-selectin is stored in endothelial granules and moves rapidly to the luminal surface in response to histamine and thrombin.	Abbas 11e p.45	*“is stored in cytoplasmic granules of endothelial cells and is rapidly redistributed to the luminal surface in response to histamine from mast cells”*		
M4-F005	IL-1 and TNF induce E-selectin expression on endothelial cells within 1 to 2 hours.	Abbas 11e p.45	*“E-selectin is synthesized and expressed on the endothelial cell surface within 1 to 2 hours in response to the cytokines interleukin-1 (IL-1)”*		
M4-F006	The best-defined leukocyte ligand for E-selectin and P-selectin is the tetrasaccharide sialyl Lewis X.	Abbas 11e p.45	*“The best defined of these is the tetrasaccharide sialyl Lewis X”*		
M4-F007	PSGL-1 on leukocytes displays sialyl Lewis X, the major carbohydrate ligand for P-selectin.	Abbas 11e p.45	*“P-selectin glycoprotein ligand 1 (PSGL-1) is posttranslationally modified to display sialyl Lewis X, which serves as the major carbohydrate ligand for P-selectin”*		
M4-F008	Endothelial Weibel-Palade bodies contain preformed P-selectin.	Janeway 8e p.105	*“induce the rapid externalization of granules called Weibel-Palade bodies in the endothelial cells. These granules contain preformed P-selectin”*		
M4-F009	Selectin-ligand bonds have low affinity and a fast off-rate, so they form and break repeatedly and the leukocyte rolls.	Abbas 11e p.49	*“the interactions of selectins with their ligands are of low affinity (Kd ~100 μm) with a fast off-rate”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F010	Integrins are heterodimers of one alpha chain and one beta chain.	Abbas 11e p.45	*“all of which are heterodimers containing 1 of 18 types of α chains and 1 of 8 types of β chains”*		
M4-F011	LFA-1 (CD11aCD18) binds ICAM-1 (CD54) on cytokine-activated endothelium.	Abbas 11e p.45	*“One important ligand for LFA-1 is intercellular adhesion molecule 1 (ICAM-1, CD54), a membrane glycoprotein expressed on cytokine-activated endothelial cells”*		
M4-F012	MAC-1 (CD11bCD18) is also complement receptor 3 (CR3); it binds ICAM-1 and iC3b-opsonized particles.	Abbas 11e p.45	*“MAC-1 (αMβ2, CD11bCD18, or complement receptor 3 [CR3]) on circulating monocytes binds to ICAM-1 and mediates adhesion to endothelium”*		
M4-F013	VLA-4 binds VCAM-1 (CD106) on cytokine-activated endothelium.	Abbas 11e p.45	*“VLA-4 binds to vascular cell adhesion molecule 1 (VCAM-1, CD106), an Ig superfamily protein expressed on cytokine-activated endothelial cells”*		
M4-F014	Chemokine receptor signals change integrin conformation to a high-affinity state; this is called inside-out signalling.	Abbas 11e p.46	*“alter the binding functions of the extracellular domain of integrins is called inside-out signaling”*		
M4-F015	Endothelial heparan sulfate proteoglycans bind chemokines and display them to adherent leukocytes.	Abbas 11e p.48	*“Chemokines produced in the tissues bind to heparan sulfate proteoglycans on endothelial cells that line postcapillary venules”*		
M4-F016	Leukocyte recruitment follows selectin-mediated rolling, chemokine-mediated integrin activation, stable integrin-mediated arrest and transmigration.	Abbas 11e p.49	*“Selectin-mediated rolling of leukocytes on endothelium... Chemokine-mediated increase in affinity of integrins... Stable integrin-mediated arrest... Transmigration of leukocytes through the endothelium”*		
M4-F017	For firm arrest, cytokines upregulate endothelial VCAM-1 (binds VLA-4) and ICAM-1 (binds LFA-1 and MAC-1).	Abbas 11e p.49	*“These ligands include VCAM-1, which binds the integrin VLA-4, and ICAM-1, which binds LFA-1 and MAC-1 integrins”*		
M4-F018	Leukocytes usually migrate between endothelial cells; Abbas calls this paracellular transmigration or diapedesis.	Abbas 11e p.49	*“Leukocytes usually migrate out of blood vessels between endothelial cells, a process called paracellular transmigration or diapedesis”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F019	Transmigration involves CD31 and a reversible disruption of the VE-cadherin junction complex.	Abbas 11e p.50	*“notably CD31... This process requires a transient and reversible disruption of adherens junction proteins, primarily the VE-cadherin complex”*		
M4-F020	Janeway uses the term diapedesis for the movement through the basement membrane, after the cell squeezes between endothelial cells using PECAM (CD31).	Janeway 8e p.106	*“The movement through the basement membrane is known as diapedesis, and it enables phagocytes to enter the subendothelial tissues.”*		
M4-F021	Chemotaxis is movement of cells along a chemoattractant concentration gradient toward its source.	Abbas 11e p.46	*“Chemokines stimulate movement of cells along the concentration gradient of the secreted proteins toward their source, a process called chemotaxis”*		
M4-F022	CXC chemokines with an ELR motif support neutrophil migration.	Abbas 11e p.46	*“CXC chemokines with ELR motifs support neutrophil migration.”*		
M4-F023	Chemokine receptors are seven-transmembrane G protein-coupled receptors (GPCRs).	Abbas 11e p.47	*“The receptors for chemokines belong to the seven-transmembrane, guanosine triphosphate-binding (G) protein-coupled receptor (GPCR) superfamily.”*		
M4-F024	Neutrophils express CXCR1 and CXCR2, which bind CXCL1 and CXCL8 (IL-8).	Abbas 11e p.50	*“Neutrophils express CXCR1 and CXCR2, which bind CXCL1 and CXCL8 (IL-8), the major chemokines with ELR motifs”*		
M4-F025	Classical monocytes express CCR2; its most important ligand for monocyte recruitment is CCL2 (MCP-1).	Abbas 11e p.50	*“classical monocytes... express CCR2. This receptor binds several chemokines, the most important one for monocyte recruitment being CCL2 (MCP-1)”*		
M4-F026	C5a, C3a and leukotriene B4 (LTB4) are chemoattractants for phagocytes and act through GPCRs.	Abbas 11e p.48	*“The C3a and C5a complement fragments... and the arachidonic acid metabolite leukotriene B4 (LTB4) are important chemoattractants that promote phagocyte migration”*		
M4-F027	Formyl peptide receptor 1 (FPR1) recognizes N-formylmethionyl bacterial peptides, which are among the most potent leukocyte chemoattractants.	Abbas 11e p.79	*“The bacterial peptide ligands that bind this receptor are some of the most potent known chemoattractants for leukocytes.”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F028	The fMLP receptor is a GPCR; its signals direct cell movement and induce reactive oxygen species in the phagolysosome.	Janeway 8e p.79	*“Signaling through the fMLP receptor also induces the production of microbicidal reactive oxygen species in the phagolysosome.”*		
M4-F029	Neutrophils exposed to C5a or CXCL8 are activated to produce the respiratory burst.	Janeway 8e p.103	*“neutrophils exposed to C5a or CXCL8 are activated to produce the respiratory burst that generates oxygen radicals and nitric oxide”*		
M4-F030	Neutrophils are usually the most numerous leukocytes in tissue within 24 to 48 hours of infection and are then replaced by monocytes.	Abbas 11e p.50	*“neutrophils are usually the most numerous type of leukocyte to accumulate in tissues, within 24 to 48 hours after the onset of an infection”*		
M4-F031	Janeway states that neutrophil influx usually peaks within the first 6 hours of an inflammatory response.	Janeway 8e p.106	*“Their influx usually peaks within the first 6 hours of an inflammatory response, whereas monocytes can be recruited later, through the action of CCL2.”*		
M4-F032	An adult makes more than 1 x 10^11 neutrophils per day; each circulates from a few hours to up to 5 days.	Abbas 11e p.14	*“produces more than 1 × 10^11 neutrophils per day, each of which circulates in the blood from a few hours to up to 5 days”*		
M4-F033	Neutrophil life span in tissues is 1 to 2 days; tissue-resident macrophages can live for years.	Abbas 11e p.15	*“Life span in tissues 1–2 days ... Tissue-resident macrophages: years”*		
M4-F034	Neutrophil production is stimulated by G-CSF and GM-CSF.	Abbas 11e p.14	*“Production of neutrophils is stimulated by granulocyte colony-stimulating factor (G-CSF) and granulocyte-macrophage colony-stimulating factor (GM-CSF).”*		
M4-F035	Neutrophil specific granules contain lysozyme, collagenase and elastase; azurophilic granules contain myeloperoxidase, defensins and cathelicidins.	Abbas 11e p.14	*“specific granules, are filled with enzymes, such as lysozyme, collagenase, and elastase... azurophilic granules... contain enzymes (e.g., myeloperoxidase)”*		
M4-F036	Dead and dying neutrophils are a major component of pus.	Janeway 8e p.82	*“Dead and dying neutrophils are a major component of the pus that forms in abscesses”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F037	Phagocytosis is an active, energy-dependent engulfment of particles larger than 0.5 micrometre.	Abbas 11e p.95	*“Phagocytosis is an active, energy-dependent process of engulfment of large particles (greater than 0.5 μm in diameter) into vesicles.”*		
M4-F038	Binding of the microbe to phagocyte receptors is the first step of phagocytosis; some are PRRs such as C-type lectins and scavenger receptors.	Abbas 11e p.95	*“binding of microbes to these receptors is the first step in phagocytosis... pattern recognition receptors, including C-type lectins and scavenger receptors”*		
M4-F039	Opsonization is the coating of a pathogen with antibodies and/or complement proteins so that phagocytes take it up more readily.	Janeway 8e p.49	*“Opsonization refers to the coating of a pathogen by antibodies and/or complement proteins so that it is more readily taken up”*		
M4-F040	FcγRI (CD64) is a high-affinity IgG receptor on macrophages and neutrophils; it binds IgG1 and IgG3 and mediates phagocytosis.	Abbas 11e p.297	*“FcγRI (CD64) High (Kd ~10–9 M); binds IgG1 and IgG3, can bind monomeric IgG ... Phagocytosis; activation of phagocytes”*		
M4-F041	CR1 (CD35) is a high-affinity receptor for C3b and C4b; phagocytes use it to bind and internalize opsonized particles.	Abbas 11e p.307	*“CR1 is a high-affinity receptor for C3b and C4b... Phagocytes use this receptor to bind and internalize particles opsonized with C3b or C4b.”*		
M4-F042	CR1 alone is inefficient at inducing phagocytosis; simultaneous IgG-FcγR binding or IFN-γ activation enhances it.	Abbas 11e p.313	*“By itself, CR1 is inefficient at inducing the phagocytosis of C3b-coated microbes, but its ability to do so is enhanced”*		
M4-F043	C5a can activate macrophages to ingest bacteria bound to CR1.	Janeway 8e p.63	*“the small complement fragment C5a can activate macrophages to ingest bacteria bound to their CR1 receptors”*		
M4-F044	CR3 (MAC-1, CD11bCD18) is an integrin receptor for iC3b; its beta chain CD18 is shared with LFA-1 and CR4.	Abbas 11e p.309	*“The type 3 complement receptor, also called MAC-1 (CR3, CD11bCD18), is an integrin that functions as a receptor for the iC3b fragment”*		
M4-F045	CR3 also directly recognizes and phagocytoses microbes that bear beta-glucans.	Janeway 8e p.78	*“The complement receptor CR3 also directly recognizes and phagocytoses microbes bearing B-glucans.”*		
M4-F046	The membrane cup closes around the particle to form a phagosome, which fuses with a lysosome.	Abbas 11e p.95	*“the top of the cup closes over and pinches off the interior of the cup to form an inside-out intracellular vesicle”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F047	Fusion of phagosomes with lysosomes forms phagolysosomes, where most microbicidal mechanisms are concentrated.	Abbas 11e p.95	*“The fusion of phagocytic vacuoles (phagosomes) with lysosomes results in the formation of phagolysosomes, where most of the microbicidal mechanisms are concentrated.”*		
M4-F048	Signals from TLRs, Fc and C3b receptors, IFN-γ receptors and CD40 activate phagocytes to kill ingested microbes.	Abbas 11e p.95	*“pattern recognition receptors (such as TLRs), opsonin receptors (such as Fc and C3b receptors), receptors for cytokines (mainly IFN-γ), and CD40”*		
M4-F049	The three most important classes of microbicidal molecules are reactive oxygen species, nitric oxide and proteolytic enzymes.	Abbas 11e p.95-96	*“Three classes of microbicidal molecules are known to be the most important.”*		
M4-F050	Phagocyte (NADPH) oxidase is a multisubunit enzyme assembled mainly in the phagolysosomal membrane of activated phagocytes.	Abbas 11e p.96	*“Phagocyte oxidase is a multisubunit enzyme that is assembled in activated phagocytes, mainly in the phagolysosomal membrane.”*		
M4-F051	Phagocyte oxidase reduces molecular oxygen to superoxide, with NADPH as cofactor.	Abbas 11e p.96	*“to chemically reduce molecular oxygen into ROS, such as superoxide radicals, with the reduced form of nicotinamide adenine dinucleotide phosphate (NADPH) acting as a cofactor”*		
M4-F052	Superoxide is dismutated to hydrogen peroxide; myeloperoxidase uses hydrogen peroxide to convert halide ions into toxic hypohalous acids.	Abbas 11e p.96	*“Superoxide is enzymatically dismutated into hydrogen peroxide, which is used by the enzyme myeloperoxidase to convert normally unreactive halide ions into reactive hypohalous acids”*		
M4-F053	The process is called the respiratory burst because it requires oxygen consumption.	Abbas 11e p.96	*“The process by which ROS are produced is called the respiratory burst because it requires oxygen consumption”*		
M4-F054	Cytochrome b558 subunits (gp91 and p22) sit in secondary granule membranes; p40, p47 and p67 are cytosolic in resting neutrophils.	Janeway 8e p.81	*“subunits (gp91 and gp22) of the NADPH oxidase are localized in the membranes of secondary granules; the other oxidase components (p40, p47, and p67)”*		
M4-F055	fMLP and C5a receptor signals activate Rac2, which is needed for NADPH oxidase assembly.	Janeway 8e p.80	*“activating Ras-family GTPases, such as Rac2, that are necessary for the assembly of the active NADPH oxidase”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F056	Superoxide dismutase converts superoxide to hydrogen peroxide; further reactions produce hypochlorite and hydroxyl radical.	Janeway 8e p.81	*“O2- is converted by superoxide dismutase (SOD) to hydrogen peroxide (H2O2)... to microbicidal hypochlorite (OCl-) and the hydroxyl (•OH) radical”*		
M4-F057	Potassium and hydrogen ions enter the phagolysosome after superoxide forms; this releases and activates granule enzymes such as cathepsin G and elastase.	Janeway 8e p.81	*“Potassium and hydrogen ions are then drawn into the phagolysosome... Acidification dissociates granule enzymes such as cathepsin G and elastase”*		
M4-F058	Janeway notes that ROS alone may not kill; ROS-driven K+ influx raises vacuole pH to the optimum for microbicidal proteins.	Janeway 8e p.536	*“ROS cause an influx of K+ ions into the phagocytic vacuole, increasing the pH to the optimal level for the action of microbicidal peptides”*		
M4-F059	Inducible nitric oxide synthase (iNOS) is absent in resting macrophages and is induced by TLR signals, especially with IFN-γ.	Abbas 11e p.96	*“absent in resting macrophages but is induced in response to microbial products that activate TLRs, especially in combination with IFN-γ”*		
M4-F060	iNOS converts arginine to citrulline and releases NO; NO combines with H2O2 or superoxide to form peroxynitrite.	Abbas 11e p.96	*“iNOS catalyzes the conversion of arginine to citrulline and freely diffusible NO gas is released... to produce highly reactive peroxynitrite radicals”*		
M4-F061	Neutrophils make low or no nitric oxide; macrophages make it after iNOS induction.	Abbas 11e p.15	*“Nitric oxide Low levels or none Induced after transcriptional activation of iNOS”*		
M4-F062	Elastase and cathepsin G are important neutrophil proteolytic enzymes for bacterial killing.	Abbas 11e p.96	*“One of the important enzymes in neutrophils is elastase, a broad-spectrum serine protease... Another important enzyme is cathepsin G.”*		
M4-F063	Defensins are cationic peptides made by epithelia and by neutrophils; they kill microbes mainly by disrupting microbial membranes.	Abbas 11e p.80	*“Defensins are produced by epithelial cells of mucosal surfaces and by granule-containing leukocytes, including neutrophils”*		
M4-F064	Neutrophil alpha-defensins are stored in primary granules, which fuse with phagosomes.	Janeway 8e p.46	*“generating a mature cationic defensin that is stored in the so-called primary granules”*		
M4-F065	Lysozyme digests cell walls of some Gram-positive bacteria; lactoferrin sequesters iron.	Janeway 8e p.80	*“Lysozyme: digests cell walls of some Gram-positive bacteria ... Lactoferrin (sequesters Fe2+)”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F066	NETs are extruded DNA and histone strands with bound granule proteins (lysozyme, elastase, defensins) that trap and kill bacteria and fungi.	Abbas 11e p.96	*“are composed of strands of DNA and histones to which high concentrations of antimicrobial granule contents are bound, including lysozyme, elastase, and defensins”*		
M4-F067	NET formation needs PAD4-mediated histone citrullination, elastase, myeloperoxidase and phagocyte oxidase.	Abbas 11e p.96	*“NET formation requires citrullination of histones by the enzyme peptidylarginine deiminase (PAD4), as well as neutrophil serine protease elastase, myeloperoxidase, and phagocyte oxidase”*		
M4-F068	NET extrusion usually causes neutrophil death (NETosis); excessive NET formation contributes to autoimmune and inflammatory disease.	Abbas 11e p.96	*“usually leads to neutrophil cell death, referred to as NETosis... excessive NET formation contributes to autoimmune and other inflammatory diseases”*		
M4-F069	Phagocyte defects cause invasive bacterial or fungal infections of skin, liver and respiratory tract, mainly Aspergillus and Candida for fungi.	Abbas 11e p.486	*“invasive infections of the skin, liver, and respiratory tract with bacteria or fungi, the latter predominantly involving Aspergillus and Candida species”*		
M4-F070	Janeway groups phagocyte defects into defects of production, adhesion, activation and killing.	Janeway 8e p.533	*“deficiencies in phagocyte production, phagocyte adhesion, phagocyte activation, and phagocyte killing of microorganisms”*		
M4-F071	Flow cytometry can detect absence of a cell-surface or intracellular protein in immunodeficiency work-up.	Abbas 11e p.486	*“flow cytometric analysis of immune cells to evaluate numbers and relative proportions of these cells or the absence of a cell surface or intracellular protein”*		
M4-F072	CGD is caused by mutations in components of NADPH oxidase (phagocyte oxidase).	Abbas 11e p.486	*“Chronic granulomatous disease (CGD) is caused by mutations in components of the NADPH oxidase (also called phagocyte oxidase) enzyme complex.”*		
M4-F073	About two-thirds of CGD cases are X-linked recessive (gp91, the 91-kD subunit of cytochrome b558); the rest are autosomal recessive.	Abbas 11e p.486	*“Approximately two-thirds of cases show an X-linked recessive pattern of inheritance, and the remainder are autosomal recessive.”*		
M4-F074	CGD affects about 1 in 200,000 individuals in the United States.	Abbas 11e p.486	*“estimated to affect approximately 1 in 200,000 individuals in the United States”*	⚠	

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F075	Autosomal recessive CGD (other oxidase subunits) can be milder, with later onset.	Janeway 8e p.82	*“Autosomal mutations in other subunits of the NADPH oxidase can also cause chronic granulomatous disease, but can be milder and have a later onset.”*		
M4-F076	Invasive Aspergillus infection is the leading cause of death in CGD.	Abbas 11e p.487	*“Invasive infection with the fungus Aspergillus is the leading cause of death.”*		
M4-F077	Catalase-producing organisms such as Staphylococcus are troublesome in CGD because catalase destroys the hydrogen peroxide from residual host superoxide.	Abbas 11e p.487	*“such as Staphylococcus, produce catalase, which destroys the microbicidal hydrogen peroxide that may be produced by host cells from residual superoxide”*		
M4-F078	In CGD, uncontrolled infection drives T cell-mediated macrophage activation and granuloma formation.	Abbas 11e p.487	*“they stimulate chronic cell-mediated immune responses, resulting in T cell-mediated macrophage activation and the formation of granulomas”*		
M4-F079	CGD management includes antibiotic and antifungal prophylaxis, IFN-γ, and consideration of HSCT; gene therapy trials are ongoing.	Abbas 11e p.487	*“Treatment with the cytokine IFN-γ can be used in addition to preventive antibiotic and antifungal therapy... standard of care to consider hematopoietic stem cell transplantation”*	⚠	
M4-F080	After HSCT, restoring neutrophil superoxide production to at least 15% of normal greatly improves resistance to infection.	Abbas 11e p.487	*“If neutrophil superoxide production is restored to at least 15% of normal levels after HSCT, resistance to infection is greatly improved.”*	⚠	
M4-F081	Secondary HLH can complicate CGD.	Abbas 11e p.498	*“Secondary HLH can also be seen as a complication of CGD”*		
M4-F082	Myeloperoxidase deficiency impairs intracellular killing and gives a CGD-like but less severe phenotype.	Janeway 8e p.536	*“Deficiencies in glucose-6-phosphate dehydrogenase and myeloperoxidase also impair intracellular bacterial killing and lead to a similar, although less severe, phenotype.”*		
M4-F083	Janeway lists G6PD deficiency and myeloperoxidase deficiency, with CGD, as defects of the respiratory burst.	Janeway 8e p.534	*“The respiratory burst is defective in chronic granulomatous disease, glucose-6-phosphate dehydrogenase (G6PD) deficiency, and myeloperoxidase deficiency.”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F084	LADs are autosomal recessive; they cause failure of neutrophil recruitment, severe early bacterial infections and inability to make pus.	Abbas 11e p.488	*“The leukocyte adhesion deficiencies are a group of autosomal recessive disorders... severe bacterial infections starting early in life, and an inability to make pus”*		
M4-F085	LAD-1 is caused by CD18 mutations with absent or reduced beta2 integrins (LFA-1, MAC-1, p150,95).	Abbas 11e p.488	*“absent or reduced expression of the β2 integrins... as a result of mutations in the CD18 gene”*		
M4-F086	LAD-1 causes recurrent bacterial and fungal infections, impaired wound healing and common leukocytosis.	Abbas 11e p.488	*“characterized by recurrent bacterial and fungal infections and impaired wound healing. Leukocytosis is common”*		
M4-F087	In LAD-1, adherence, aggregation, chemotaxis, phagocytosis and cytotoxicity of leukocytes are defective.	Abbas 11e p.488	*“including adherence to endothelium, neutrophil aggregation and chemotaxis, phagocytosis, and cytotoxicity mediated by neutrophils, NK cells, and T lymphocytes”*		
M4-F088	LAD-2 is caused by a defective GDP-fucose transporter; sialyl Lewis X is absent and leukocytes do not roll.	Abbas 11e p.488	*“caused by a mutation in a guanosine diphosphate (GDP)-fucose transporter... the absence of leukocyte rolling”*		
M4-F089	LAD-2 also causes severe developmental abnormalities and the Bombay blood group phenotype; it is a congenital disorder of glycosylation.	Abbas 11e p.488	*“This abnormality in fucosylation seen in LAD-2 also contributes to a Bombay blood group phenotype”*		
M4-F090	LAD-3 is caused mainly by KINDLIN-3 mutations that block inside-out chemokine-induced integrin activation.	Abbas 11e p.488	*“This deficiency is caused by a defect in the inside-out signaling pathway that mediates chemokine-induced integrin activation”*		
M4-F091	LAD-3 causes a life-threatening bleeding disorder from defective platelet aggregation, with normal platelet counts.	Abbas 11e p.488	*“a life-threatening bleeding disorder that requires blood transfusions because of defective platelet aggregation, even though blood platelet counts are normal”*		
M4-F092	Infections in LAD resist antibiotic treatment and show impaired wound healing and no pus.	Janeway 8e p.535	*“impaired wound healing and, in pyogenic bacterial infections, the absence of pus formation. The infections that occur in these patients are resistant to antibiotic treatment”*		
M4-F093	Chédiak-Higashi syndrome (CHS) is autosomal recessive with pyogenic infections, partial oculocutaneous albinism and giant lysosomes in leukocytes.	Abbas 11e p.488	*“recurrent infections by pyogenic bacteria, partial oculocutaneous albinism... The neutrophils, monocytes, and lymphocytes of these patients contain giant lysosomes.”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F094	CHS is caused by mutations in LYST, which regulates intracellular trafficking of lysosomes.	Abbas 11e p.488	*“caused by mutations in the gene encoding the protein LYST, which regulates intracellular trafficking of lysosomes”*		
M4-F095	In CHS, NK cell function is impaired and leukocytes show defective chemotaxis and phagocytosis.	Abbas 11e p.488	*“NK cell function in these patients is impaired... Leukocytes are also defective in chemotaxis and phagocytosis”*		
M4-F096	Untransplanted CHS patients usually develop hemophagocytic lymphohistiocytosis (HLH) with fever, DIC, organomegaly and hemophagocytosis.	Abbas 11e p.488	*“CHS patients who are not transplanted will likely develop a life-threatening syndrome that includes fever, disseminated intravascular coagulation”*	⚠	
M4-F097	In CHS, phagosomes and lysosomes fail to fuse properly; phagocytes have enlarged granules and poor intracellular killing.	Janeway 8e p.536	*“causes a failure of lysosomes and phagosomes to fuse properly; the phagocytes in these patients have enlarged granules and impaired intracellular killing ability”*		
M4-F098	CHS features metallic silver hair, poor vision and platelet dysfunction with bleeding.	Janeway 8e p.540	*“The hair is typically a metallic silver color, vision is poor because of abnormalities in retinal pigment cells, and platelet dysfunction causes increased bleeding.”*		
M4-F099	Severe congenital neutropenia (SCN, Kostmann disease) has a persistent neutrophil count below 0.5 x 10^9/L.	Janeway 8e p.533	*“the neutrophil count is persistently less than 0.5 x 10^9 per liter of blood”*		
M4-F100	Cyclic neutropenia has neutrophil counts that fluctuate with a cycle of about 21 days.	Janeway 8e p.533	*“neutrophil numbers that fluctuate from near normal to very low or undetectable, with an approximate cycle time of 21 days”*		
M4-F101	Mutations in ELA2 (neutrophil elastase) are the most common cause of SCN; some ELA2 mutations cause cyclic neutropenia.	Janeway 8e p.534	*“sporadic or autosomal dominant mutations of the gene that encodes neutrophil elastase (ELA2)... Some mutations of ELA2 cause cyclic neutropenia”*		
M4-F102	HAX1 deficiency causes an autosomal recessive form of SCN.	Janeway 8e p.534	*“Autosomal recessive forms of SCN have also been identified. Deficiency of the mitochondrial protein HAX1 leads to increased apoptosis”*		
M4-F103	SCN and cyclic neutropenia carry an increased risk of myelodysplasia or myeloid leukemia; G-CSF is the mainstay of treatment.	Janeway 8e p.534	*“increased risk of developing myelodysplasia or myeloid leukemia... G-CSF which is now the mainstay of treatment for congenital neutropenias”*	⚠	

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F104	MSMD is severe disease from weakly virulent mycobacteria and other intracellular pathogens such as Salmonella; BCG vaccine can disseminate.	Abbas 11e p.498	*“predisposed to severe disease caused by weakly virulent mycobacteria that do not cause disease in healthy individuals... the vaccine can cause disseminated mycobacterial infection”*		
M4-F105	MSMD genes include IL-12p40, IL-12Rbeta1, both IFN-γ receptor chains, STAT1 and NEMO (IKKγ).	Abbas 11e p.498	*“Mutations in the genes encoding IL-12p40, the IL-12Rβ1 chain, and both chains of the IFN-γ receptor, as well as some mutations in STAT1 and IKKγ”*		
M4-F106	IL-12 from DCs and macrophages drives IFN-γ production by T cells and NK cells.	Abbas 11e p.498	*“IL-12R signaling stimulates the synthesis of IFN-γ by helper T cells, cytotoxic T cells, and NK cells”*		
M4-F107	Because p40 is shared by IL-12 and IL-23, p40 deficiency causes both IL-12 and IL-23 deficiency.	Janeway 8e p.538	*“p40 is shared by IL-12 and IL-23, and so p40 deficiency can cause both IL-12 and IL-23 deficiency.”*		
M4-F108	Total absence of IFN-γ receptor expression markedly reduces granuloma formation.	Janeway 8e p.538	*“In the total absence of IFN-γ receptor expression, granuloma formation is much reduced”*		
M4-F109	MyD88 or IRAK4 deficiency causes severe pyogenic infections with little inflammation ('cold' infections).	Janeway 8e p.535	*“recurrent, severe peripheral, and invasive infections by pyogenic bacteria that elicit little inflammation, a situation known as a 'cold' infection”*		
M4-F110	Genetic testing is often used relatively early in the diagnosis of inborn errors of immunity.	Abbas 11e p.485	*“genetics is often used relatively early in diagnosis, compared to other clinical disciplines”*		
M4-F111	Janeway places neutrophil elastase (ELA2) in the azurophilic granules.	Janeway 8e p.534	*“encodes neutrophil elastase (ELA2), a component of the azurophilic granules”*		
M4-F112	Janeway states that neutrophil cathelicidin propeptides are stored in secondary granules and are activated by elastase released from primary granules.	Janeway 8e p.46	*“the cathelicidin propeptides are stored in the secondary granules... encounter neutrophil elastase that has been released from primary granules”*		
M4-F113	Janeway describes four steps of extravasation: selectin rolling, tight integrin binding, crossing the endothelium and basement membrane, and migration through tissue along chemokine gradients.	Janeway 8e p.106	*“The migration of leukocytes out of blood vessels, the process known as extravasation, is thought to occur in four steps.”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M4-F114	In CGD, loss of NADPH oxidase activity causes defective superoxide production, so phagocytes fail to kill ingested microbes.	Abbas 11e p.486-487	*“The loss of NADPH oxidase activity results in defective production of superoxide anion... results in a failure to kill phagocytosed microbes.”*		
M5-F001	Complement is a major effector mechanism of humoral immunity and also an important effector mechanism of innate immunity.	Abbas 11e p.300	*“one of the major effector mechanisms of humoral immunity and is also an important effector mechanism of innate immunity”*		
M5-F002	Complement proteases circulate as inactive zymogens and become active only after proteolytic cleavage.	Janeway 8e p.49	*“synthesized as inactive pro-enzymes, or zymogens, which only become enzymatically active after proteolytic cleavage”*		
M5-F003	Each activated complement protease cleaves many molecules of the next zymogen, so the signal amplifies at each step.	Janeway 8e p.49	*“each becoming an active protease that cleaves and activates many molecules of the next zymogen in the pathway”*		
M5-F004	The larger cleavage fragment gets the suffix b and the smaller fragment gets the suffix a; C2 is the one exception.	Janeway 8e p.49	*“The larger fragment is always designated by the suffix b, with one exception.”*		
M5-F005	For C2, the larger fragment that stays attached is called C2a and the released smaller fragment is called C2b.	Abbas 11e p.305	*“because the attached, larger fragment is called the a piece and the released part is the b fragment”*		
M5-F006	The proteins are numbered in order of discovery, not order of action; the classical pathway reaction sequence is C1, C4, C2, C3, C5, C6, C7, C8 and C9.	Janeway 8e p.49	*“named in the order of their discovery rather than the sequence of reactions. The reaction sequence in the classical pathway”*		
M5-F007	Three pathways activate complement: the classical pathway (antibody bound to antigen), the alternative pathway (microbial surfaces without antibody) and the lectin pathway (lectins that bind microbial carbohydrates).	Abbas 11e p.301	*“the classical pathway, which is activated by certain classes of antibodies bound to antigens; the alternative pathway”*		
M5-F008	The alternative and lectin pathways are innate effector mechanisms; the classical pathway is a major mechanism of adaptive humoral immunity.	Abbas 11e p.301	*“The alternative and lectin pathways are effector mechanisms of innate immunity, whereas the classical pathway is a major mechanism of adaptive humoral immunity.”*		
M5-F009	All three pathways can start without antibody, as part of innate immunity.	Janeway 8e p.71	*“All three pathways can be initiated independently of antibody as part of innate immunity.”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M5-F010	The central event of complement activation is cleavage of C3 by a C3 convertase into C3a and C3b, with covalent attachment of C3b to the target surface.	Abbas 11e p.301	*“The central event in complement activation is proteolysis of the complement protein C3”*		
M5-F011	Cleavage of C3 exposes a reactive thioester bond in C3b; if C3b does not bond to a nearby surface, the thioester is hydrolyzed and C3b becomes inactive.	Janeway 8e p.52	*“If no bond is made, the thioester is rapidly hydrolyzed, inactivating C3b.”*		
M5-F012	Regulatory proteins on normal host cells inhibit complement activation; microbes lack these proteins, so activation proceeds on microbial surfaces.	Abbas 11e p.301	*“Complement activation is inhibited by regulatory proteins that are present on normal host cells and absent from microbes.”*		
M5-F013	The classical pathway starts when C1 binds the CH2 domain of IgG or the CH3 domain of IgM that has bound antigen.	Abbas 11e p.305	*“binding of the complement protein C1 to the CH2 domains of IgG or the CH3 domains of IgM”*		
M5-F014	Among human IgG subclasses, IgG1 and IgG3 are the most efficient complement activators.	Abbas 11e p.305	*“Among IgG antibodies, IgG1 and IgG3 (in humans) are more efficient activators of complement than are other subclasses.”*		
M5-F015	IgG2 activates complement weakly and IgG4 does not activate complement.	Abbas 11e p.305	*“IgG2 has some ability to activate complement, but IgG4 does not.”*		
M5-F016	C1 consists of C1q, which binds antibody, and C1r and C1s, which are serine proteases.	Abbas 11e p.305	*“C1q binds to the antibody, and C1r and C1s are proteases”*		
M5-F017	Each C1q must bind at least two Ig heavy chains to be activated, so only antigen-bound antibody, not free antibody, activates the classical pathway.	Abbas 11e p.305	*“each C1q molecule must bind to at least two Ig heavy chains to be activated”*		
M5-F018	One antigen-bound IgM molecule can bind two or more C1q molecules, so IgM fixes complement more efficiently than IgG.	Abbas 11e p.305	*“a single molecule of IgM can bind two or more C1q molecules”*		
M5-F019	C1q also binds pentraxins such as C-reactive protein (CRP) and serum amyloid P, and it binds apoptotic bodies.	Abbas 11e p.305	*“C1q also binds to pentraxins like C-reactive protein and serum amyloid protein, and can also bind to apoptotic bodies”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M5-F020	Binding of C1q activates C1r; C1r cleaves and activates C1s; C1s cleaves C4, and C4b binds covalently to the surface through its thioester bond.	Abbas 11e p.305	*“enzymatic activation of the associated C1r, which cleaves and activates C1s”*		
M5-F021	C2 binds surface C4b and is cleaved by C1s; the larger C2a fragment stays bound to C4b.	Abbas 11e p.305	*“a larger C2a fragment that remains physically associated with C4b on the cell surface”*		
M5-F022	C4b2a is the classical pathway C3 convertase.	Abbas 11e p.305	*“The resulting C4b2a complex is the classical pathway C3 convertase”*		
M5-F023	C3b binds to C4b2a to form C4b2a3b, the classical pathway C5 convertase.	Abbas 11e p.306	*“form a C4b2a3b complex. This complex functions as the classical pathway C5 convertase”*		
M5-F024	C3 is homologous to C4, and factor B is homologous to C2.	Abbas 11e p.306	*“C3 in the alternative pathway is homologous to C4 in the classical pathway, and Factor B is homologous to C2.”*		
M5-F025	The lectin pathway starts when plasma mannose-binding lectin (MBL) or ficolins bind microbial polysaccharides.	Abbas 11e p.306	*“triggered by the binding of microbial polysaccharides to circulating lectins, such as plasma mannose-binding lectin (MBL), or to ficolins”*		
M5-F026	MBL binds mannose residues on microbial polysaccharides.	Abbas 11e p.306	*“MBL binds to mannose residues on polysaccharides”*		
M5-F027	Ficolins (L-, H- and M-ficolin) bind N-acetylglucosamine-containing glycans through a fibrinogen-like domain.	Abbas 11e p.306	*“the fibrinogen-like domain of ficolin binds N-acetylglucosamine–containing glycans”*		
M5-F028	MASP1 may activate MASP2, and MASP2 cleaves C4 and C2.	Abbas 11e p.307	*“MASP1 may activate MASP2 and MASP2 is the protease that cleaves C4 and C2”*		
M5-F029	The lectin pathway forms the same C3 convertase as the classical pathway, C4b2a.	Janeway 8e p.54	*“forming C4b2a, which is the C3 convertase of the lectin pathway”*		
M5-F030	Plasma C3 is cleaved spontaneously at a low rate of 1-2% of total C3 per hour (C3 tick-over).	Abbas 11e p.301	*“continuously cleaved at a low rate (1%–2% of the total plasma C3 per hour)”*		
M5-F031	Tick-over gives C3(H2O), which binds factor B; factor D then cleaves factor B and a short-lived fluid-phase C3 convertase, C3(H2O)Bb, forms.	Janeway 8e p.58	*“producing a short-lived fluid-phase C3 convertase, C3(H,O)Bb”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M5-F032	Surface C3b binds factor B, factor D cleaves factor B (Ba is released, Bb stays), and C3bBb forms the alternative pathway C3 convertase.	Abbas 11e p.301	*“The C3bBb complex is the alternative pathway C3 convertase”*		
M5-F033	The alternative pathway C3 convertase amplifies complement activation that starts by the alternative, classical or lectin pathway.	Abbas 11e p.301	*“the alternative pathway C3 convertase functions to amplify complement activation when it is initiated by the alternative, classical, or lectin pathway”*		
M5-F034	Factor D is the only activating protease of complement that circulates as an active enzyme, not as a zymogen.	Janeway 8e p.61	*“the only activating protease of the complement system to circulate as an active enzyme rather than a zymogen”*		
M5-F035	Properdin binds and stabilizes C3bBb, and it attaches preferentially to microbial surfaces.	Abbas 11e p.302	*“properdin, can bind to and stabilize the C3bBb complex, and the attachment of properdin is favored on microbial as opposed to normal host cells”*		
M5-F036	The alternative pathway C5 convertase contains one Bb and two C3b molecules (C3bBb3b).	Abbas 11e p.303	*“a complex containing one Bb moiety and two molecules of C3b, which functions as the alternative pathway C5 convertase”*		
M5-F037	Janeway writes the C5 convertases as C4b2a3b (classical and lectin) and C3b2Bb (alternative).	Janeway 8e p.62	*“C3b binds both to C4b2a and C3bBb, forming the active C5 convertases C4b2a3b and C3b,Bb”*		
M5-F038	C5 binds to the C3b part of a C5 convertase and is cleaved by C2a or Bb.	Janeway 8e p.62	*“rendered susceptible to cleavage by the serine protease activity of C2a or Bb”*		
M5-F039	Unlike C3b and C4b, C5b does not bind covalently to the cell surface.	Janeway 8e p.62	*“Unlike C3b and C4b, C5b is not covalently bound to the cell surface.”*		
M5-F040	C5b binds C6 and then C7; the C5b-7 complex inserts into the lipid bilayer.	Abbas 11e p.307	*“C7 from the plasma then binds to the α chain of C5b and forms the C5b-C6-C7 (C5b-7) complex”*		
M5-F041	C8 binds to form C5b-8, which makes small, unstable pores of 0.4-3 nm.	Abbas 11e p.307	*“forms unstable pores that range from 0.4 to 3 nm in diameter”*		
M5-F042	Recruitment of the first C9 molecule is the rate-limiting step of MAC formation.	Abbas 11e p.307	*“The rate-limiting step in MAC formation is the recruitment of the first C9 molecule”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M5-F043	Abbas states that the MAC pore contains C5b, C6, C7, C8 and up to 18 C9 molecules.	Abbas 11e p.307	*“C5b-9 complexes containing C5b, C6, C7, C8, and up to 18 molecules of C9”*		
M5-F044	Janeway states that 10-16 C9 molecules polymerize to form the MAC.	Janeway 8e p.66	*“the polymerization of 10-16 molecules of C9”*		
M5-F045	Water enters through the MAC pore, and the cell swells and ruptures (osmotic lysis).	Abbas 11e p.307	*“The entry of water results in osmotic swelling and rupture of the cells”*		
M5-F046	MAC-mediated lysis is critical for defense against only a few pathogens, such as thin-walled Neisseria.	Abbas 11e p.314	*“such as bacteria of the genus Neisseria, which have thin cell walls”*		
M5-F047	Deficiency of C5-C9 is associated with susceptibility only to Neisseria species.	Janeway 8e p.66	*“deficiencies in complement components C5-C9 have been associated with susceptibility only to Neisseria species”*		
M5-F048	C3b and C4b bind CR1; iC3b binds CR3 (MAC-1) and CR4.	Abbas 11e p.313	*“C3b and C4b (the latter generated by the classical pathway only) bind to CR1, and iC3b binds to CR3 (MAC-1) and CR4”*		
M5-F049	CR1 alone triggers phagocytosis of C3b-coated microbes inefficiently; simultaneous IgG binding to Fc gamma receptors enhances it.	Abbas 11e p.313	*“By itself, CR1 is inefficient at inducing the phagocytosis of C3b-coated microbes”*		
M5-F050	Binding of iC3b to CR3 is sufficient on its own to stimulate phagocytosis.	Janeway 8e p.64	*“the binding of iC3b to the receptor CR3 is sufficient on its own to stimulate phagocytosis”*		
M5-F051	People without a spleen are susceptible to disseminated pneumococcal and meningococcal septicemia.	Abbas 11e p.313	*“are susceptible to disseminated pneumococcal and meningococcal septicemia”*		
M5-F052	C5a, C4a and C3a bind to mast cells and cause degranulation with release of vasoactive mediators such as histamine.	Abbas 11e p.313	*“All three peptides bind to mast cells and induce degranulation, with the release of vasoactive mediators, such as histamine.”*		
M5-F053	C3a, C4a and C5a are called anaphylatoxins because the mast cell reactions they trigger are characteristic of anaphylaxis.	Abbas 11e p.313	*“These peptides are also called anaphylatoxins because the mast cell reactions they trigger are characteristic of anaphylaxis”*		
M5-F054	C5a is the most potent mediator of mast cell degranulation; C3a is about 20-fold and C4a about 2500-fold less potent.	Abbas 11e p.314	*“C5a is the most potent mediator of mast cell degranulation, C3a is approximately 20-fold less potent, and C4a is*		

ID	Fact	Source	Supporting quote	Vol.	Verified
			approximately 2500-fold less”		
M5-F055	In neutrophils, C5a stimulates motility, firm adhesion to endothelium and, at high doses, the respiratory burst.	Abbas 11e p.313	*“C5a stimulates motility, firm adhesion to endothelial cells, and, at high doses, stimulation of the respiratory burst”*		
M5-F056	Large systemic amounts of anaphylatoxins cause circulatory collapse, a shock-like syndrome similar to a systemic IgE-mediated allergic reaction.	Janeway 8e p.64	*“they induce a generalized circulatory collapse, producing a shock-like syndrome similar to that seen in a systemic allergic reaction”*		
M5-F057	C3a and C5a can degranulate mast cells without allergen cross-linking of the high-affinity IgE receptor.	Abbas 11e p.470	*“The complement fragments C3a and C5a can cause mast cell degranulation, and this is why they were named anaphylatoxins.”*		
M5-F058	C5a is a chemoattractant for eosinophils.	Abbas 11e p.473	*“the complement product C5a and the lipid mediators PAF and LTB4 produced by mast cells also function as chemoattractants for eosinophils”*		
M5-F059	The C5a receptor C5aR1 is a G protein-coupled receptor on neutrophils, eosinophils, basophils, monocytes, macrophages, mast cells and other cells.	Abbas 11e p.309	*“C5aR1, is the most thoroughly characterized. It is a member of the G protein–coupled receptor family”*		
M5-F060	Plasma carboxypeptidases remove the C-terminal arginine of C3a and C5a; the des-Arg forms keep only about 10% of the activity.	Abbas 11e p.312	*“C3a des-Arg and C5a des-Arg, which each have only approximately 10% of the activity of the native forms”*		
M5-F061	CR1 on erythrocytes binds circulating immune complexes that carry C3b and C4b and carries them to the liver and spleen for removal by phagocytes.	Abbas 11e p.307	*“CR1 on erythrocytes binds circulating immune complexes with attached C3b and C4b”*		
M5-F062	CR2 (CD21) binds C3d, C3dg and iC3b; on B cells it forms a complex with CD19 and CD81.	Abbas 11e p.308	*“a trimolecular complex that includes two other noncovalently attached proteins called CD19 and CD81”*		
M5-F063	CR2 is the receptor for Epstein-Barr virus (EBV).	Abbas 11e p.310	*“Receptor for EBV”*		
M5-F064	C3d on an antigen binds CR2 on B cells and facilitates B-cell activation.	Abbas 11e p.314	*“The C3d protein generated from C3 binds to CR2 on B cells and facilitates B-cell activation”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M5-F065	CR3 (MAC-1, CD11b/CD18) is an integrin that binds iC3b and also ICAM-1 on endothelium.	Abbas 11e p.309	*“The type 3 complement receptor, also called MAC-1 (CR3, CD11bCD18), is an integrin”*		
M5-F066	CR4 (CD11c/CD18) also binds iC3b; CD11c is a marker of dendritic cells.	Abbas 11e p.309	*“CD11c is abundantly expressed on dendritic cells and is used as a marker for this cell type.”*		
M5-F067	Mutations in the shared CD18 chain remove CR3 and CR4 and cause leukocyte adhesion deficiency (LAD).	Abbas 11e p.315	*“The disease caused by this gene defect is called leukocyte adhesion deficiency”*		
M5-F068	C1 inhibitor (C1-INH) is a serpin; active C1r2s2 cleaves it, it binds covalently, and C1r2s2 dissociates from C1q.	Abbas 11e p.311	*“C1 INH is a serine protease inhibitor (serpin) that mimics the normal substrates of C1r and C1s”*		
M5-F069	C1-INH also inactivates MASP2 and so dampens the lectin pathway.	Abbas 11e p.311	*“by inactivating MASP2, C1-INH also dampens the lectin pathway”*		
M5-F070	C1-INH also inhibits kallikrein and coagulation factor XII, which promote bradykinin formation.	Abbas 11e p.311	*“C1-INH is an inhibitor of other plasma serine proteases besides C1, including kallikrein and coagulation factor XII”*		
M5-F071	Hereditary angioedema (HAE) is an autosomal dominant disease caused by deficiency of C1-INH.	Abbas 11e p.311	*“An autosomal dominant inherited disease called hereditary angioedema is due to a deficiency of C1-INH.”*		
M5-F072	The edema mediators in HAE include C2 kinin (a C2 fragment) and bradykinin.	Abbas 11e p.311	*“The mediators of edema formation in patients with hereditary angioedema include a proteolytic fragment of C2, called C2 kinin, and bradykinin.”*		
M5-F073	Factor H binds C3b, displaces Bb and is a cofactor for factor I-mediated cleavage of C3b.	Abbas 11e p.310	*“Binds C3b and displaces Bb”*		
M5-F074	Factor H blocks only Bb binding to C3b, so it regulates the alternative pathway but not the classical pathway.	Abbas 11e p.311	*“Factor H inhibits binding of only Bb to C3b and is thus a regulator of the alternative but not the classical pathway.”*		
M5-F075	Sialic acid-rich host cell surfaces favor binding of factor H over factor B.	Abbas 11e p.311	*“cell surfaces rich in sialic acid favor binding of the regulatory protein Factor H over the alternative pathway protein Factor B”*		
M5-F076	C4-binding protein (C4BP) is a plasma protein that binds C4b, displaces C2 and is a cofactor for factor I-mediated cleavage of C4b.	Abbas 11e p.310	*“Binds C4b and displaces C2”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M5-F077	Factor I is a plasma serine protease that cleaves C3b and C4b, with factor H, MCP, C4BP or CR1 as cofactor.	Abbas 11e p.310	*“cleaves C3b and C4b by using factor H, MCP, C4BP, or CR1 as cofactors”*		
M5-F078	Factor I circulates in active form but cleaves C3b and C4b only when they are bound to a cofactor protein.	Janeway 8e p.68	*“it circulates in active form but can cleave C3b and C4b only when they are bound to a membrane cofactor protein”*		
M5-F079	Factor I cleavage of C3b generates iC3b, C3d and C3dg; these do not take part in activation but bind receptors on phagocytes and B cells.	Abbas 11e p.312	*“Factor I–mediated cleavage of C3b generates the fragments called iC3b, C3d, and C3dg”*		
M5-F080	Membrane cofactor protein (MCP, CD46) is on leukocytes, epithelial cells and endothelial cells and is a cofactor for factor I cleavage of C3b and C4b.	Abbas 11e p.310	*“Cofactor for factor I–mediated cleavage of C3b and C4b”*		
M5-F081	Decay-accelerating factor (DAF, CD55) displaces C2a from C4b and Bb from C3b, so it dissociates both C3 convertases.	Abbas 11e p.310	*“Displaces C2a from C4b and Bb from C3b (dissociation of C3 convertases)”*		
M5-F082	CR1, DAF and factor H displace Bb from C3b; CR1, MCP and factor H are cofactors for factor I cleavage of C3b.	Janeway 8e p.60	*“CR1, DAF, and factor H displace Bb from C3b, and CR1, MCP, and factor H catalyze the cleavage of bound C3b”*		
M5-F083	CD59 is a GPI-linked membrane protein that incorporates into the assembling MAC after C5b-8 insertion and blocks addition of C9.	Abbas 11e p.312	*“It works by incorporating itself into assembling MACs after the membrane insertion of C5b-8, thereby inhibiting the subsequent addition of C9 molecules”*		
M5-F084	CD59 is also called protectin.	Janeway 8e p.70	*“CD59 or protectin, which inhibits the binding of C9 to the C5b678 complex”*		
M5-F085	Plasma S protein binds soluble C5b-7 complexes and prevents their insertion into bystander cell membranes.	Abbas 11e p.312	*“plasma proteins such as S protein, which functions by binding to soluble C5b,6,7 complexes”*		
M5-F086	S protein is also called vitronectin.	Janeway 8e p.70	*“vitronectin, also known as S-protein, bind to the C5b67 complex”*		
M5-F087	Failure to make GPI anchors removes DAF and CD59 from blood cells and causes paroxysmal nocturnal hemoglobinuria (PNH).	Abbas 11e p.311	*“results in the failure to express many GPI-linked membrane proteins, including DAF and CD59”*		
M5-F088	Factor I deficiency depletes plasma C3 through unregulated fluid-phase C3 convertase formation and causes pyogenic infections.	Abbas 11e p.315	*“In patients with Factor I deficiency, plasma C3 is depleted”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M5-F089	In complement-mediated hemolytic-uremic syndrome, the most common mutations are in the factor H gene; MCP mutations also occur.	Abbas 11e p.315	*“the most common mutations in this condition are in the Factor H gene”*		
M5-F090	Some microbes, such as Gram-positive bacteria and some fungi, have thick cell walls that prevent binding of complement proteins such as the MAC.	Abbas 11e p.316	*“Some microbes express thick cell walls that prevent the binding of complement proteins, such as the MAC.”*		
M5-F091	Many pathogens evade complement by recruiting host regulators, for example factor H through surface sialic acid.	Abbas 11e p.316	*“Microbes can evade the complement system by recruiting host complement regulatory proteins.”*		
M5-F092	Janeway states that iC3b is a ligand for CR2, CR3 and CR4, but not for CR1.	Janeway 8e p.64	*“producing iC3b, which is a ligand for the complement receptors CR2, CR3 and CR4, but not CR1”*		
M5-F093	Abbas lists the CR1 ligands as C3b > C4b > iC3b.	Abbas 11e p.310	*“C3b > C4b > iC3b”*		
M5-F094	In mouse studies of acute bacterial infection, alternative pathway activation releases C5a, which directly triggers mast cell degranulation.	Abbas 11e p.482	*“Studies in mice have indicated that mast cells can be activated by IgE-independent mechanisms in the course of an acute bacterial infection”*		
M5-F095	IgM against capsular polysaccharides activates the classical pathway and causes phagocytic clearance of pneumococci and meningococci in the spleen.	Abbas 11e p.313	*“activate the classical pathway of complement, and cause phagocytic clearance of the bacteria in the spleen”*		
M5-F096	The most important action of complement is to help phagocytes take up and destroy pathogens.	Janeway 8e p.62	*“The most important action of complement is to facilitate the uptake and destruction of pathogens by phagocytic cells.”*		
M5-F097	Regulators act at three levels: they inhibit C3 convertase formation, break down C3 and C5 convertases, and inhibit MAC formation.	Abbas 11e p.311	*“Different regulatory mechanisms inhibit the formation of C3 convertases in the early steps of complement activation, break down and inactivate C3 and C5 convertases”*		
M5-F098	The reactive thioester of C3b and C4b cannot distinguish a host cell surface from a pathogen surface.	Janeway 8e p.68	*“has no mechanism for distinguishing an acceptor hydroxyl or amine group on a host cell from a similar group on the surface of a pathogen”*		
M5-F099	People deficient in C3 or in molecules that catalyze C3b deposition are susceptible to infection by a wide range of extracellular bacteria.	Janeway 8e p.64	*“individuals deficient in C3 or in molecules that catalyze C3b deposition show an increased susceptibility to infection by a*		

ID	Fact	Source	Supporting quote	Vol.	Verified
			wide range of extracellular bacteria”		
M6-F001	C2 deficiency is the most common human complement deficiency.	Abbas 11e p.315	*“C2 deficiency is the most common human complement deficiency”*		
M6-F002	More than 50% of patients with C1q, C2 or C4 deficiency develop systemic lupus erythematosus (SLE).	Abbas 11e p.315	*“More than 50% of patients with C1q, C2, and C4 deficiencies develop systemic lupus erythematosus”*		
M6-F003	The SLE association may come from inadequate clearance of circulating immune complexes; uncleared complexes deposit in vessel walls and tissues and cause inflammation.	Abbas 11e p.315	*“it may be related to inadequate clearance of circulating immune complexes because of defects in complement activation”*		
M6-F004	Complement may also help to clear apoptotic bodies, which are likely sources of the nuclear antigens in lupus.	Abbas 11e p.315	*“Complement may also play an important role in the clearance of apoptotic bodies containing fragmented DNA”*		
M6-F005	Some patients with C2 or C4 deficiency have more infections; others have no symptoms.	Abbas 11e p.315	*“Some patients with C2 or C4 deficiency show increased susceptibility to infections, and others are asymptomatic”*		
M6-F006	Defects in the early classical pathway components (C1, C4, C2) mainly impair processing of immune complexes and clearance of apoptotic cells, which causes immune-complex disease.	Janeway 8e p.532	*“Defects in the early components of the classical pathway predominantly affect the processing of immune complexes and the clearance of apoptotic cells”*		
M6-F007	Genetic deficiencies of C1q, C1r, C4, C2 and C3 are described in humans.	Abbas 11e p.315	*“Genetic deficiencies in classical pathway components, including C1q, C1r, C4, C2, and C3, have been described”*		
M6-F008	C3 deficiency causes frequent serious pyogenic bacterial infections that can be fatal, which shows the central role of C3 in opsonization and phagocytosis.	Abbas 11e p.315	*“Deficiency of C3 is associated with frequent serious pyogenic bacterial infections that may be fatal”*		
M6-F009	Defects in C3 or in C3 activation increase susceptibility to pyogenic bacteria, including Streptococcus pneumoniae.	Janeway 8e p.532	*“increased susceptibility to infection by a range of pyogenic bacteria, including S. pneumoniae”*		
M6-F010	Janeway lists C3 deficiency as causing pyogenic and Neisseria infections and sometimes immune-complex disease.	Janeway 8e p.532	*“Deficiency leads to infection with C3 pyogenic bacteria and Neisseria spp. Sometimes immune-complex disease”*		
M6-F011	MBL deficiency is relatively common (about 5% of the population).	Janeway 8e p.533	*“are relatively common (5% of the population)”*	⚠	

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F012	MBL deficiency may cause a mild immunodeficiency with more bacterial infections in early childhood; MASP2 defects give a similar phenotype.	Janeway 8e p.533	*“MBL deficiency may be associated with a mild immunodeficiency with an increased incidence of bacterial infection in early childhood”*		
M6-F013	Deficiencies of alternative pathway components increase susceptibility to meningococcal infections.	Abbas 11e p.315	*“Deficiencies in components of the alternative pathway result in increased susceptibility to meningococcal infections”*		
M6-F014	Factor B and factor D deficiencies are rare; X-linked recessive properdin deficiency is more common.	Abbas 11e p.315	*“Factor B and Factor D deficiencies are rare, but X-linked recessive properdin deficiency is more common”*		
M6-F015	Defects in factor D and properdin give a Neisseria susceptibility similar to that of terminal component defects.	Janeway 8e p.532	*“A similar susceptibility phenotype is found in patients with defects in the alternative complement pathway components, factor D and properdin”*		
M6-F016	Properdin-deficient patients are particularly susceptible to Neisseria meningitidis.	Janeway 8e p.59	*“Properdin-deficient patients are particularly susceptible to infections with Neisseria meningitidis”*		
M6-F017	In C5, C6, C7, C8 or C9 deficiency the only consistent clinical problem is disseminated Neisseria infection, because MAC lysis is important against these thin-walled bacteria.	Abbas 11e p.315	*“the only consistent clinical problem in these patients is a propensity for disseminated infections by Neisseria bacteria”*		
M6-F018	Defects in the membrane attack components C5-C9 cause susceptibility almost only to Neisseria species.	Janeway 8e p.532	*“result almost exclusively in susceptibility to Neisseria species”*		
M6-F019	In a Japanese population study, the yearly risk of N. meningitidis infection was about 1 in 200 with an inherited MAC protein deficiency versus 1 in 2,000,000 normally (10,000-fold).	Janeway 8e p.533	*“a risk of 1 in 200 to a person in the same population with an inherited deficiency of one of the membrane-attack complex proteins”*	⚠	
M6-F020	In factor I deficiency, unregulated fluid-phase C3 convertase forms by tickover and depletes plasma C3.	Abbas 11e p.315	*“plasma C3 is depleted as a result of the unregulated formation of fluid-phase C3 convertase”*		
M6-F021	The clinical result of factor I deficiency is more infections with pyogenic bacteria.	Abbas 11e p.315	*“The clinical consequence is increased infections with pyogenic bacteria”*		
M6-F022	Homozygous factor I deficiency causes a de facto C3 deficiency.	Janeway 8e p.533	*“leading to a de facto C3 deficiency”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F023	Factor H inhibits only the binding of Bb to C3b, so it regulates the alternative pathway but not the classical pathway.	Abbas 11e p.311	*“Factor H inhibits binding of only Bb to C3b and is thus a regulator of the alternative but not the classi-”*		
M6-F024	Factor H deficiency is rare and causes excess alternative pathway activation, C3 consumption and glomerulonephritis.	Abbas 11e p.315	*“Factor H deficiency is rare and is characterized by excess alternative pathway activation, consumption of C3, and glomerulonephritis”*		
M6-F025	C3 nephritic factor (C3NeF) is an autoantibody that stabilizes C3bBb against factor H and causes C3 consumption; patients often have glomerulonephritis.	Abbas 11e p.315	*“C3NeF stabilizes C3bBb and protects the complex from Factor H–mediated dissociation, which results in unregulated consumption of C3”*		
M6-F026	Atypical haemolytic uraemic syndrome (aHUS) involves defective complement regulation; the most common mutations are in the factor H gene, and the MCP gene is also mutated in many patients.	Abbas 11e p.315	*“the most common mutations in this condition are in the Factor H gene”*		
M6-F027	In aHUS, children present with microangiopathic haemolytic anaemia, thrombocytopenia and acute renal failure; mutant factor H or MCP binds C3b/C4b on endothelium less well.	Abbas 11e p.315	*“children present with microangiopathic hemolytic anemia, thrombocytopenia, and acute renal failure”*		
M6-F028	Deficiencies in MCP, factor I or factor H can cause aHUS; heterozygous mutations are enough to predispose.	Janeway 8e p.70	*“individuals heterozygous for mutations in the regulatory proteins MCP factor I, or factor H”*		
M6-F029	CD59 is GPI-linked and blocks the addition of C9 to C5b-8, which prevents MAC formation on host cells.	Abbas 11e p.312	*“thereby inhibiting the subsequent addition of C9 molecules”*		
M6-F030	Mutations in PIG-A cause paroxysmal nocturnal haemoglobinuria (PNH) through defective GPI anchors for CD59 and DAF (CD55).	Abbas 11e p.315	*“as a result of defective GPI anchors for CD59 and DAF”*		
M6-F031	PNH shows recurrent intravascular haemolysis, which leads to chronic haemolytic anaemia and venous thrombosis.	Abbas 11e p.312	*“Recurrent intravascular hemolysis in turn leads to chronic hemolytic anemia and venous thrombosis”*		
M6-F032	The PNH mutation is not inherited; it is an acquired somatic mutation in haematopoietic stem cells.	Abbas 11e p.312	*“is not inherited but is an acquired somatic mutation in hematopoietic stem cells”*		
M6-F033	Anti-C5 antibodies are approved for PNH and atypical HUS.	Abbas 11e p.316	*“approved for use in patients with paroxysmal nocturnal hemoglobinuria, in patients with the atypical hemolytic uremic syndrome”*	⚠	

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F034	C1-INH is a serpin that inhibits C1r, C1s and MASP2, so it controls the classical and lectin pathways.	Abbas 11e p.311	*“The proteolytic activity of C1r, C1s, and MASP2 is inhibited by a plasma protein called C1 inhibitor (C1-INH)”*		
M6-F035	Abbas names C2 kinin (a C2 fragment) and bradykinin as mediators of edema in HAE.	Abbas 11e p.311	*“include a proteolytic fragment of C2, called C2 kinin, and bradykinin”*		
M6-F036	C1-INH also inhibits kallikrein and coagulation factor XII, which both promote bradykinin formation.	Abbas 11e p.311	*“including kallikrein and coagu-”*		
M6-F037	In HAE, uncontrolled spontaneous complement activation produces excess cleaved C4 and C2 fragments.	Janeway 8e p.67	*“chronic spontaneous complement activation leads to the production of excess cleaved fragments of C4 and C2”*		
M6-F038	In HAE, C1-INH protein can fall below 20-30% of normal, so C1 activation is not controlled and C4 and C2 breakdown increases.	Abbas 11e p.311	*“are sufficiently reduced (<20%–30% of normal) that activa-”*		
M6-F039	Angioedema can be mediated by bradykinin and/or mast cell mediators including histamine.	WAO/EAACI HAE GL 2017 p.5	*“Angioedema can be mediated by bradykinin and/or mast cell mediators including histamine”*		
M6-F040	HAE-1 (C1-INH deficiency) has low antigenic and low functional C1-INH levels.	WAO/EAACI HAE GL 2017 p.5	*“characterized by low antigenic and functional C1-INH levels”*		
M6-F041	HAE-2 (C1-INH dysfunction) has normal or elevated antigenic but low functional C1-INH levels.	WAO/EAACI HAE GL 2017 p.5	*“characterized by normal (or elevated) antigenic but low functional C1-INH levels”*		
M6-F042	HAE with normal C1-INH (HAE nC1-INH) includes HAE-FXII, HAE-ANGPT1, HAE-PLG and HAE of unknown mutation (HAE-UNK).	WAO/EAACI HAE GL 2017 p.6	*“HAE nC1-INH, hereditary angioedema with normal C1-inhibitor levels, either due to a mutation in FXII”*		
M6-F043	Acquired angioedema with low C1-INH (AAE-C1-INH) is angioedema from C1-INH deficiency on an acquired basis.	WAO/EAACI HAE GL 2017 p.5	*“Acquired angioedema with low C1-inhibitor, henceforth called AAE-C1-INH”*		
M6-F044	ACE inhibitor-induced angioedema (ACEI-AE) is an acquired bradykinin-mediated angioedema with normal C1-INH.	WAO/EAACI HAE GL 2017	*“these may be bradykinin-mediated (eg, ACE inhibitor-induced”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
		p.6			
M6-F045	The 2017 guideline recommendations apply only to HAE-1/2.	WAO/EAACI HAE GL 2017 p.6	*“This guideline only refers to HAE-1/2.”*		
M6-F046	HAE nC1-INH is very rare and clinically resembles HAE-1/2; bradykinin seems to play a major role mainly in patients with an FXII mutation.	WAO/EAACI HAE GL 2017 p.6	*“Its clinical appearance largely resembles that of”*		
M6-F047	In most patients with HAE nC1-INH no gene mutation is found.	WAO/EAACI HAE GL 2017 p.6	*“no gene mutation can be found”*		
M6-F048	HAE-1/2 is a rare autosomal dominant condition affecting about 1 in 50,000 people.	WAO/EAACI HAE GL 2017 p.6	*“HAE-1/2 is a rare autosomal dominant condition affecting an esti-”*	⚠	
M6-F049	HAE-1/2 is caused by one of more than 450 mutations in SERPING1, the gene for C1-INH.	WAO/EAACI HAE GL 2017 p.6	*“mutations in the SERPING1 gene, which codes for C1-INH”*	⚠	
M6-F050	In about 20-25% of patients a de novo SERPING1 mutation causes the disease.	WAO/EAACI HAE GL 2017 p.6	*“In approximately 20%-25% of patients, a de novo mutation of SERP-”*	⚠	
M6-F051	C1-INH is the major inhibitor of C1r, C1s, MASP1/2, plasma kallikrein and factor XIIa, and a minor inhibitor of plasmin.	WAO/EAACI HAE GL 2017 p.6	*“C1-INH is a serine protease inhibitor (SERPIN) and the major inhi-”*		
M6-F052	Bradykinin is the primary mediator of swelling in HAE-1/2.	WAO/EAACI HAE GL 2017 p.6	*“The primary mediator of swelling in HAE-1/2 is bradykinin.”*		
M6-F053	Bradykinin is a nonapeptide made when active plasma kallikrein cleaves high-molecular-weight kininogen (HMWK).	WAO/EAACI HAE GL 2017 p.6	*“ated when active plasma kallikrein cleaves high-molecular-weight”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F054	Factor XII autoactivates on negatively charged surfaces and converts prekallikrein to plasma kallikrein; C1-INH inhibits both.	WAO/EAACI HAE GL 2017 p.6	*“by the protease factor XII, which can easily autoactivate upon con-”*		
M6-F055	Metalloproteases including angiotensin-converting enzyme (ACE) rapidly degrade bradykinin.	WAO/EAACI HAE GL 2017 p.6	*“Bradykinin is rapidly metabolized by endogenous metalloproteases including angiotensin-converting enzyme (ACE).”*		
M6-F056	Bradykinin increases vascular permeability mainly through the bradykinin B2 receptor.	WAO/EAACI HAE GL 2017 p.6	*“mediated through the bradykinin B2 receptor”*		
M6-F057	Suspect HAE-1/2 with recurrent angioedema plus: family history, childhood/adolescent onset, painful abdominal attacks, upper airway edema, no response to antihistamines, glucocorticoids or epinephrine, prodromes, no wheals.	WAO/EAACI HAE GL 2017 p.6	*“nes, glucocorticoids, or epinephrine, (6) presence of prodromal signs or”*		
M6-F058	A family history may be absent in up to 25% of HAE-1/2 patients.	WAO/EAACI HAE GL 2017 p.6	*“be present in up to 25% of patients”*		
M6-F059	Recommendation 1: test all suspected patients for C1-INH function, C1-INH protein and C4, and repeat abnormal low results to confirm.	WAO/EAACI HAE GL 2017 p.6	*“are assessed for blood levels of C1-INH function, C1-”*	⚠	
M6-F060	HAE-1 is about 85% of patients; C4 is usually low in HAE-1/2 but has limited sensitivity and specificity.	WAO/EAACI HAE GL 2017 p.6	*“C4 levels are usually low in HAE-1/”*		
M6-F061	C3 and CH50 are expected to be normal in HAE and testing them is usually not helpful.	WAO/EAACI HAE GL 2017 p.6-7	*“Complement C3 and CH50”*		
M6-F062	SERPING1 sequencing can support diagnosis (including prenatal), but biochemical C1-INH testing is effective and cheaper; sequencing can miss some mutations.	WAO/EAACI HAE GL 2017 p.7	*“biochemical C1-INH testing is effective”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F063	AAE-C1-INH has the same basic lab profile as HAE-1 but later onset, underlying lymphoma or MGUS, occasional B symptoms and often low C1q.	WAO/EAACI HAE GL 2017 p.7	*“oratory profile (C1-INH function, C1-INH protein, and C4) is indistin-”*		
M6-F064	Measure C1q especially in new-onset angioedema after age 40 with a negative family history.	WAO/EAACI HAE GL 2017 p.7	*“new onset of angioedema after the age of 40 years and a negative”*		
M6-F065	C1q is nearly always normal in HAE and low in 75% of AAE-C1-INH; it may be normal in AAE patients taking anabolic androgens.	WAO/EAACI HAE GL 2017 p.7	*“C1q is nearly always normal in HAE”*		
M6-F066	Many AAE-C1-INH patients have autoantibodies that inactivate C1-INH.	WAO/EAACI HAE GL 2017 p.7	*“Many patients with AAE-C1-INH have autoantibodies that inactivate C1-INH”*		
M6-F067	Wheals do not exclude HAE, because chronic spontaneous urticaria (CSU) can coexist; absence of wheals does not exclude mast cell-mediated angioedema.	WAO/EAACI HAE GL 2017 p.7	*“does not necessarily exclude HAE”*		
M6-F068	Mast cell-mediated angioedema is far more common than HAE-1/2; when the diagnosis is open and the history does not fit HAE, treat with antihistamines and, if needed, epinephrine and corticosteroids.	WAO/EAACI HAE GL 2017 p.7	*“Because mast cell-mediated angioe-”*		
M6-F069	Symptoms usually start in childhood or adolescence; median age of onset is about 12 years; subcutaneous edema is the most common and earliest symptom.	WAO/EAACI HAE GL 2017 p.10	*“The median age of symptom onset is”*		
M6-F070	Erythema marginatum is a prodromal sign in 42-58% of paediatric cases and is often mistaken for urticaria.	WAO/EAACI HAE GL 2017 p.11	*“It has been observed in 42%-”*		
M6-F071	Offspring of an HAE-1/2 patient have a 50% chance of inheriting the disease and are considered affected until tested.	WAO/EAACI HAE GL 2017 p.11	*“HAE-1/2 stands a 50% chance of inheriting the disease”*		

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F072	C4 is not useful for HAE diagnosis below age 12 months, cord-blood values are falsely low, and early tests are repeated after age 1.	WAO/EAACI HAE GL 2017 p.11	*“The measurement of C4, however, was found”*		
M6-F073	Healthy women have lower C1-INH in pregnancy, so diagnostic tests in pregnancy need caution and repeat after delivery.	WAO/EAACI HAE GL 2017 p.12	*“the plasma levels of C1-INH decrease during”*		
M6-F074	Triggers include trauma (dental, medical, surgical), oestrogen-containing contraceptives and hormone replacement, psychological stress, fatigue, febrile illness and menstruation; most attacks are unpredictable.	WAO/EAACI HAE GL 2017 p.14	*“Trauma, whether accidental or associated with dental,”*		
M6-F075	ACE inhibitors may increase or precipitate HAE attacks and must be strictly avoided; progesterone-only pills may help.	WAO/EAACI HAE GL 2017 p.14	*“should therefore be avoided strictly”*		
M6-F076	Recommendation 2: consider all attacks for on-demand treatment and treat any attack that affects or may affect the upper airway.	WAO/EAACI HAE GL 2017 p.7	*“We recommend that any attack affecting”*	⚠	
M6-F077	Recommendation 3: treat attacks as early as possible; early treatment shortens time to resolution and attack duration.	WAO/EAACI HAE GL 2017 p.7	*“We recommend that attacks are treated as early as possible.”*	⚠	
M6-F078	Recommendation 4: treat HAE attacks with C1-INH, ecallantide or icatibant; if unavailable use solvent detergent-treated plasma (SDP), then fresh frozen plasma (FFP).	WAO/EAACI HAE GL 2017 p.7-8	*“C1-INH, ecallantide, or icatibant”*	⚠	
M6-F079	Antifibrinolytics (tranexamic acid) and androgens (danazol) are not advised for on-demand treatment because they show no or minimal effect.	WAO/EAACI HAE GL 2017 p.8	*“show no or only minimal effects when used for on-demand treatment”*	⚠	
M6-F080	Recommendation 5: consider intubation or a surgical airway early in progressive upper airway edema; laryngeal attacks are emergencies.	WAO/EAACI HAE GL 2017 p.8	*“tion is considered early in progressive upper airway edema”*	⚠	

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F081	Plasma-derived C1-INH (pdC1-INH; Berinert, Cinryze) replaces the missing protein, has a half-life over 30 hours and a negligible allergy risk.	WAO/EAACI HAE GL 2017 p.8	*“The risk of allergic reactions is negligible”*	⚠	
M6-F082	Recombinant C1-INH (Ruconest) comes from transgenic rabbit milk, has a half-life of about 3 hours, is for adults and adolescents, and is contraindicated in rabbit allergy.	WAO/EAACI HAE GL 2017 p.8	*“contraindicated in patients with known or suspected allergy to rabbits”*	⚠	
M6-F083	Ecallantide is a plasma kallikrein inhibitor licensed only in the USA for patients aged 12 and older; anaphylaxis in 3-4% means a healthcare professional must give it.	WAO/EAACI HAE GL 2017 p.8	*“licensed only in the United States for the on-demand treatment”*	⚠	
M6-F084	Icatibant is a synthetic bradykinin B2 receptor antagonist for self-administered on-demand treatment in adults, with a half-life of 1-2 hours.	WAO/EAACI HAE GL 2017 p.8	*“specific and selective competitive antagonist”*	⚠	
M6-F085	Recommendation 6: all patients have on-demand medication for two attacks and carry it at all times.	WAO/EAACI HAE GL 2017 p.8	*“for on-demand treatment of two attacks and carry on-”*	⚠	
M6-F086	All C1-INH concentrates and icatibant are licensed for self-administration; product indications vary by country.	WAO/EAACI HAE GL 2017 p.7	*“icatibant are licensed for self-administration”*	⚠	
M6-F087	Procedures with mechanical impact on the upper aerodigestive tract (dental work, intubation, bronchoscopy, endoscopy) can trigger swelling, usually within 48 hours.	WAO/EAACI HAE GL 2017 p.8-9	*“Swellings associated”*		
M6-F088	Recommendation 7: give short-term prophylaxis before procedures that can induce an attack, preferably C1-INH concentrate close to the procedure (often 1000 U or 20 U/kg).	WAO/EAACI HAE GL 2017 p.9	*“We recommend short-term prophylaxis before procedures”*	⚠	
M6-F089	Attenuated androgens (5 days before, 2-3 days after) are an older alternative for short-term prophylaxis; most experts do not recommend tranexamic acid for it; breakthrough attacks can occur.	WAO/EAACI HAE GL 2017 p.9	*“is not recommended”*	⚠	

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F090	Recommendations 8-9: consider prophylaxis for high-activity life events and evaluate every patient for long-term prophylaxis at every visit.	WAO/EAACI HAE GL 2017 p.9	*“evaluated for long-term”*	⚠	
M6-F091	Recommendation 10: use C1-INH for first-line long-term prophylaxis; pdC1-INH is given twice weekly (IV, or subcutaneously at 40 or 60 U/kg).	WAO/EAACI HAE GL 2017 p.9-10	*“We recommend use of C1-inhibitor for first-line long-term”*	⚠	
M6-F092	Recommendation 11: androgens are suggested as second-line long-term prophylaxis.	WAO/EAACI HAE GL 2017 p.10	*“We suggest to use androgens as second-line long-term pro-”*	⚠	
M6-F093	Androgen side effects are common (virilization, menstrual disorders, weight gain, acne) and need semiannual blood/urine tests and yearly liver ultrasound; dose by clinical response, not by C4 or C1-INH.	WAO/EAACI HAE GL 2017 p.10	*“Virilization is the most feared complication in”*	⚠	
M6-F094	Antifibrinolytics are not recommended for long-term prophylaxis; they are used mainly when C1-INH is unavailable and androgens are contraindicated.	WAO/EAACI HAE GL 2017 p.10	*“Antifibrinolytics are not recommended for long-term prophylaxis.”*	⚠	
M6-F095	Recommendation 13: test children from HAE families as soon as possible, including all offspring of an affected parent.	WAO/EAACI HAE GL 2017 p.11	*“We recommend testing children from HAE-affected families”*	⚠	
M6-F096	Recommendation 14: use C1-INH for attacks in children under 12; the guideline states that pdC1-INH is the only approved on-demand drug in childhood.	WAO/EAACI HAE GL 2017 p.11	*“We recommend C1-INH be used for treatment of HAE”*	⚠	
M6-F097	Androgens are not recommended for long-term prophylaxis in children before Tanner stage V; tranexamic acid is preferred if C1-INH is unavailable.	WAO/EAACI HAE GL 2017 p.12	*“Androgens are not recommended for long-term prophy-”*	⚠	
M6-F098	Recommendation 15: C1-INH is the preferred therapy for HAE attacks during pregnancy and lactation; androgens are contraindicated in pregnancy because they cross the placenta.	WAO/EAACI HAE GL 2017 p.13	*“We recommend C1-INH as the preferred therapy for HAE”*	⚠	

ID	Fact	Source	Supporting quote	Vol.	Verified
M6-F099	Recommendation 16: all patients have an individual action plan and carry an HAE identification card.	WAO/EAACI HAE GL 2017 p.13	*“We recommend that all patients have an action plan.”*	⚠	
M6-F100	Recommendation 18: teach self-administration to all patients given on-demand drugs licensed for it; IV self-administration is possible from age 8.	WAO/EAACI HAE GL 2017 p.14	*“taught to self-administer”*	⚠	
M6-F101	With upper airway swelling, patients self-administer treatment and then seek emergency care, because swelling may progress or rebound.	WAO/EAACI HAE GL 2017 p.14	*“Upper airway swelling may”*		
M6-F102	Recommendation 20: screen family members (C1-INH function, C1-INH protein, C4), because the first attack may affect the airway and be fatal.	WAO/EAACI HAE GL 2017 p.15	*“We recommend that family members of individuals with”*	⚠	
M6-F103	Abbas describes HAE as autosomal dominant C1-INH deficiency with intermittent edema of skin and mucosa, abdominal pain, vomiting, diarrhoea and possible life-threatening airway obstruction.	Abbas 11e p.311	*“An autosomal dominant inherited disease called hereditary angioedema is due to a deficiency of C1-INH”*		
M6-F104	C1-INH concentrate is recommended as first-line therapy in pregnant or breastfeeding HAE-1/2 patients, and it is a safe and effective option when long-term prophylaxis is needed in pregnancy.	WAO/EAACI HAE GL 2017 p.12-13	*“C1-INH concentrate is recommended as first-line therapy for pregnant or breastfeeding HAE-1/2 patients as it is safe and effective”*	⚠	
M6-F105	Kallikrein inhibition blocks cleavage of HMWK to bradykinin and further activation of FXIIa, which stops a positive feedback loop of kallikrein production.	WAO/EAACI HAE GL 2017 p.8	*“halting the positive feedback mechanism leading to additional kallikrein production”*		
M6-F106	In 2017, rhC1-INH and ecallantide were licensed for adolescents and icatibant for children or adolescents in some countries.	WAO/EAACI HAE GL 2017 p.11	*“In 2017, icatibant was licensed for the use in children or adolescents in some countries.”*	⚠	